Compendium - Experimental methodology

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Preface

In the course on experimental methodology, students are required to plan, conduct, and report on a number of physical experiments. This compendium covers the necessary tools to carry out and analyze the experiments. The compendium is used during the theoretical lessons that precede the laboratory sessions. During the laboratory sessions, the reader is expected to be able to:

1. make appropriate assumptions about functional relationships based on measurement results
2. determine the relationships through linearization
3. complement the investigations with dimensional analysis
4. conduct an investigation using dimensionless groups

Additional instructions for report writing are provided by the teachers. Some errors may have crept into this trial edition. The teachers of experimental methodology gratefully note corrections and also tips for improvements to the compendium.

Luleå, August 1992,
Thore Bucht
Bo Kjellmert

Preface to the third edition

From the academic year 93/94, Experimental Methodology is part of the Research Methodology course at Luleå University of Technology. This compendium is written to provide some theory to assist with the laboratory sessions in Experimental Methodology. Additional literature is included for the parts on Mathematical Models and Presentation Techniques. I have tried to improve the compendium in Experimental Methodology by editing some chapters and correcting various errors.

Luleå, May 1993,
Bo Kjellmert

Preface to the fourth and fifth editions

Everything changes in this world, including this compendium. In this edition, errors from previous editions have been corrected, text has been rewritten, and chapters and text sections have been added and removed. If you have any comments or if I have misspelled, poorly formulated, or (huh!) made factual errors, please contact me at: torbj@mt.luth.se

Luleå, sunny July 1994, sunny July 1995, and sunny October 1995,
Torbjörn Löfqvist

Preface to the sixth edition

This will be the first electronic edition. Therefore, there may be some formatting errors. Additionally, I have corrected some errors and added text about polynomials, linearization of polynomials, dimensional analysis, and a cheat sheet with examples of methods that often work when practically conducting a physical experiment to find a functional relationship.

Luleå, a wonderful summer 2006,
Nils Almqvist

Preface to the first English edition

The Swedish pdf document has been converted to LateX and translated into English from the Swedish original, with big help from Copilot-AI.

Luleå, October 2024,
Corina Etz

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Chapter 1

Planning and Conducting an Experiment

1.1 Strategy

It is important to have a well-thought-out strategy for how to conduct and evaluate an experiment. The experiment must be properly planned, conducted in a suitable manner, and carefully analyzed.

1.1.1 Planning

Before starting an experiment, it is important to thoroughly analyze the problem. Which variables should be investigated? What tools are available? What equipment is needed for the measurements and how should the measurements be organized? Take your time for inventory and planning!

1.1.2 Conducting

Many experiments aim to find a functional relationship between the involved variables. The classical method to do this is to vary only one of the independent variables and measure the corresponding values of the dependent variable.

1.1.3 Analysis

Different measurement series are analyzed through curve fitting, and any relationships are complemented to a final result. How to do this is shown in the following chapters.

1.2 Tables and Diagrams

1.2.1 Tables

- A table should provide a clear overview of how quantities depend on each other.
- The table should always have a title that clarifies the content of the table.
- The table should be easy and unambiguous to read.
- Both the quantity and the unit should be specified in the table header.
- Complex tables where many variables vary become difficult to read and should be avoided.

There are different ways to write table headers. In the example below, the same time t is given in two different ways.

Table 1.1: Example of writing styles in the table header.

t (10^{-3} s)	t (ms)
5.0	5.0

In both writing styles, the time $5.0 \cdot 10^{-3}$ s = 5.0 ms is intended. Misunderstandings cannot arise. Several other writing styles occur. A writing style that cannot be misinterpreted should always be used.

1.2.2 Diagrams

- Diagrams, like tables, should always have a title that describes the content of the diagram.

- On the diagram axes, both the quantity and the unit should be indicated. As in table headers, the indication for the axes should be unambiguous.
- Measurement points are always marked with a symbol such as "i" or "•" and, where applicable, error limits are set.
- Choose the scales so that a scale division is 1, 2, or 5 length units.
- A curve that is to be fitted to a series of measurement points should be drawn as smoothly as possible. No zigzag lines! Consider any error limits, see chapter 6.

Chapter 2

Curve Fitting

2.1 Introduction

In experiments, data is usually obtained in tabular form, and one often wants to describe the relationship between the variables mathematically with a functional relationship. If one is familiar with certain elementary functions in mathematics, it is often possible to obtain the desired relationship graphically. (This can also be done numerically, e.g., using the least squares method.) In this course, we will only touch upon the graphical method, and the following example will illustrate this method.

Example 1. In an experiment, the oscillation time t for a mathematical pendulum was measured at different pendulum lengths L. The following table of measured values was obtained:

Table 2.1: Data from pendulum experiment.

Pendulum Length L (m)	Oscillation Time t (s)
0.50	1.42
0.70	1.68
0.90	1.90
1.10	2.10
1.30	2.29
1.50	2.46

We want to find a functional relationship between the two variables t and L. The most important step in the process is to plot the data in an x-y

diagram and then make a more or less qualified guess about the type of functional relationship between t and L. We will then see if the guess was correct or not. We start by plotting the data in a regular x-y diagram (making an overview curve) to make some assumption about the relationship. We realize that if the pendulum length approaches zero, the oscillation time also approaches zero, so in our table above (and in the diagram), the point $L = 0, t = 0$ should also be included. The curve we fit to the points should thus pass through the origin. We draw the curve that best fits the measurement points.

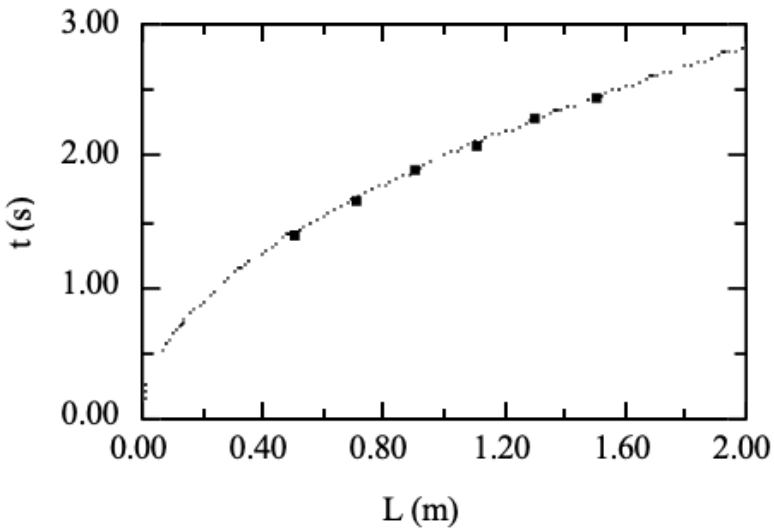


Figure 2.1: The diagram describes the pendulum's oscillation time as a function of pendulum length. The dots in the diagram denote the measurement points.

With knowledge of how elementary functions look, we can guess the functional relationship $t = g(L)$. The curve seems to illustrate some power function, and our guess is $t = aL^b$, where a and b are constants yet to be determined. To determine the constants a and b in our guessed relationship, we need to manipulate the relationship to get it into the same form as the equation of a straight line. How to do the manipulation depends on the function guessed. In our case, we guessed a power function, and to transform this into

the equation of a straight line, we take the logarithm of the relationship. We get

$$\ln(t) = \ln(aL^b)$$

Applying the logarithm laws to the right-hand side, we obtain

$$\ln(t) = \ln(a) + b\ln(L)$$

If we introduce the variables $\ln(t) = Y$, $\ln(L) = X$, and $\ln(a) = c$, we obtain a relationship of the type

$$Y = c + bX$$

i.e., a linear relationship since $\ln(a) = c$ is a constant. Comparing the equation above with the equation of a straight line, we find that the slope of the line is b and the intercept with the t-axis (the Y-axis) is $\ln(a)$. Transforming the guess into the same form as the equation of a straight line is called linearizing the relationship. We can then use the measurement data to calculate values for the constants a and b. We complete the table with values for $\ln(t)$ and $\ln(L)$.

Table 2.2: Data from pendulum experiment.

Pendulum Length L (m)	Oscillation Time t (s)	$\ln(L)$	$\ln(t)$
0.50	1.42	0.351	-0.693
0.70	1.68	0.519	-0.357
0.90	1.90	0.642	-0.105
1.10	2.10	0.742	0.095
1.30	2.29	0.829	0.262
1.50	2.46	0.900	0.405

In a coordinate system with $\ln(t)$ as the Y-coordinate and $\ln(L)$ as the X-coordinate, we draw the straight line that best fits the points. The constant b is given by the slope of the line.

We can note that it is perfectly possible to fit a straight line to the measurement data. From the diagram with $\ln(t)$ as a function of $\ln(L)$, we obtain $b \approx 0.5$ (the slope of the line). The intercept with the Y-axis, $X = \ln(L) = 0$, gives $\ln(a) \approx 0.69$, i.e., $a \approx e^{0.69} \approx 2.0$. The determination of both a and b is given by the accuracy of the reading from the diagram. Other linearizations

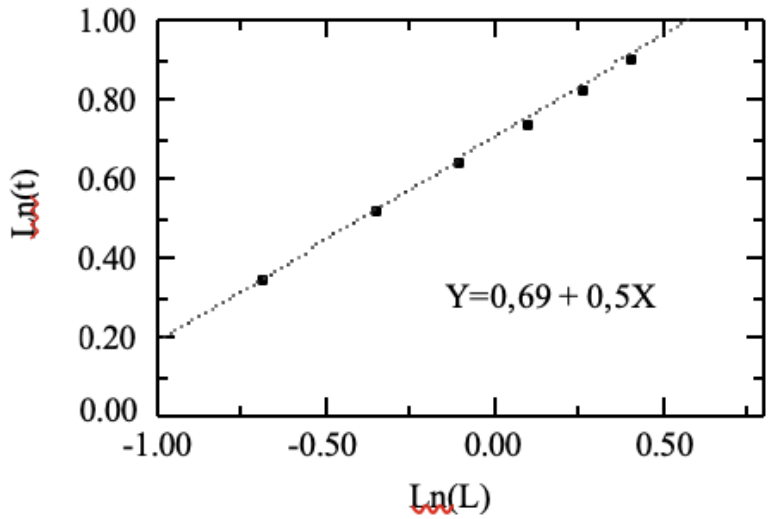


Figure 2.2: The linearized equation. The equation of the line (dashed in the figure) is written in the figure.

can also be used to determine the constant. The relationship $t = aL^{0.5}$ can, for example, be squared, and one gets $t^2 = a^2L$. If t^2 is plotted as a function of L, a straight line with the slope a^2 is obtained. In both cases, the sought relationship is $t = 2.0L^{0.5} = 2.0\sqrt{L}$. In the example above, it was possible to fit a straight line to the linearized measurement data. This means that our guessed relationship was correct. If we had guessed the wrong function and done all the steps above, it would not have been possible to fit a straight line to the measurement data. Then one has to start over from the beginning with a new guess and repeat the procedure again.

In physical contexts, all quantities have dimensions and are included with these in formulas and relationships. We will show in later chapters that exponents in most physical formulas take integer or half-integer values, e.g., -1, -0.5, 0.5, 1, etc. This can be understood from the dimensions of the quantities involved. This means that exponents in power functions can be determined with relatively high certainty from diagrams.

2.2 Choice of Approach

The choice of approach is an important step when trying to determine a functional relationship. With knowledge of the appearance of some elementary functions, it is easier to find the right relationship. Linear functions are trivial (lines are easy to recognize), so we start with power functions.

Power Functions

Power functions are functions of the type $y = ax^n$, where a and n are constants. The figure below shows three curves corresponding to different values of the constant n (with a set to one).

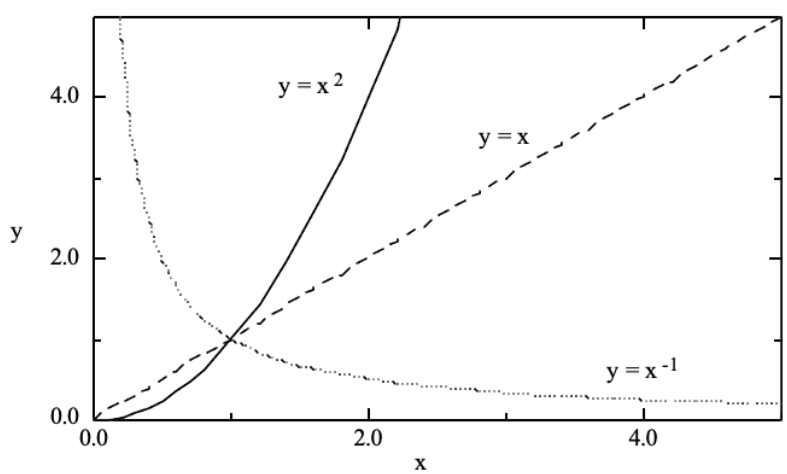


Figure 2.3: Different power functions.

We see that for $n = 1$, the function is a straight line. It can be noted that for $n < 0$, y approaches zero as x approaches infinity, and y approaches infinity as x approaches zero. In other words, the function asymptotically approaches zero for large x. There are power functions with positive n that do not pass through the origin or negative n that do not asymptotically approach zero for large x. These power functions can be written as $y = C + ax^n$. The constant C is obtained from an

overview curve when $x \to 0$ or $x \to \infty$. Many real experiments can be described with a simple factorial power function of the type:

$$y = Cx^{\alpha} z^{\beta} h^{\gamma} p^{\delta} q^{\epsilon}$$

where x, z, h, p, q are the independent variables in the experiment, C is a dimensionless constant, and the exponents $\alpha, \beta, \gamma, \delta, \epsilon$ are constants (fractions, integers).

Exponential Functions

Exponential functions are functions of the type $y = a^x$. Below are curves for different values of the constant a.

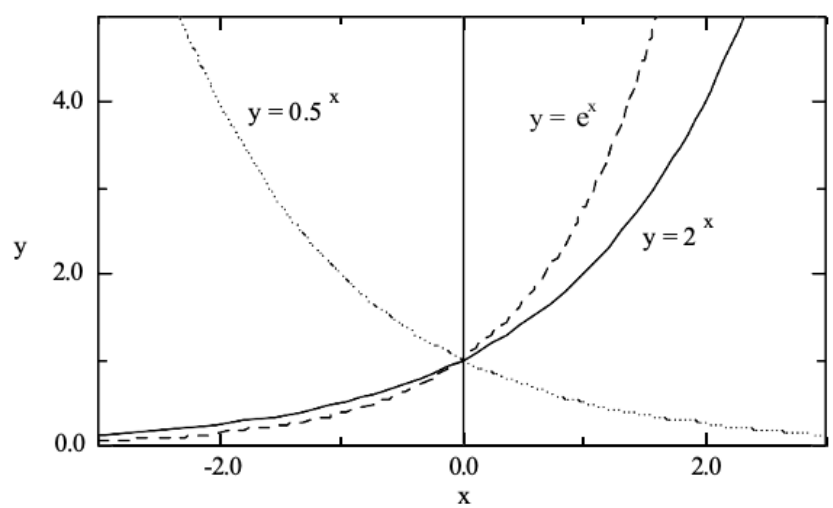


Figure 2.4: Different exponential functions.

Note that all curves pass through $y = 1$. If the measurement data does not pass through $y = 1$, it may be a function of the type $y = C + a^x$, where C and a are constants.

A Composite Function

There are many different types of functions, and we have so far only discussed the two simplest. Another type of function that can appear is composed of two different power functions, $y = ax^n + bx^m$. Below is an example where $a = b = 1$, $n = -1$, and $m = 1$.

In the figure, the solid curve is $y = x^{-1} + x$. It is composed of the two power functions $y = x^{-1}$ and $y = x$, which are plotted as dashed curves in the figure. The function is easy to recognize by its characteristic "hammock" appearance.

Polynomials

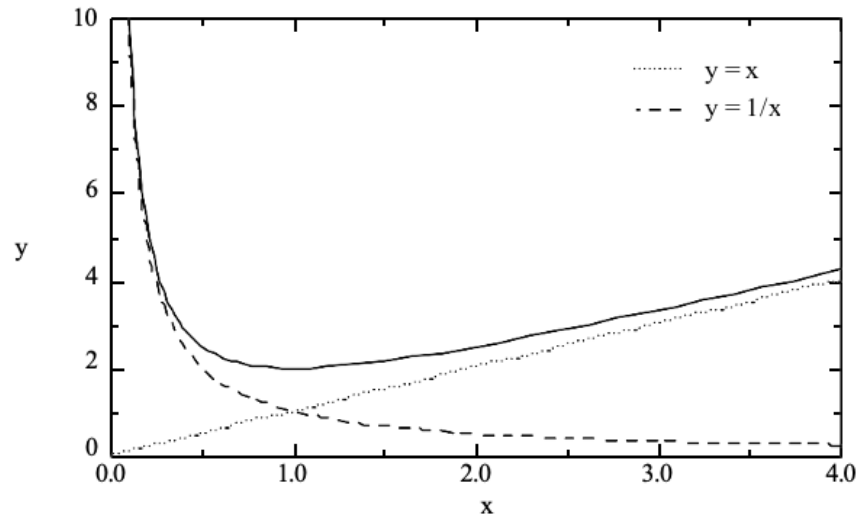


Figure 2.5: The solid curve is described by the equation $y = x^{-1} + x$.

The last type of function we discuss here is polynomials. A polynomial can be written as:

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

Polynomials can also appear as functional relationships in real experiments, although it is often more appropriate to use composite or factorial exponential functions as mathematical models for the experiments.

2.3 Linearization

Once the function type has been chosen based on an overview curve, it is time to check if the approach is correct. The simplest way to do this is to transform the approach so that it takes the same form as the equation of a straight line and then see if the linearization can be fitted to the measurement data. If the approach is correct, the fit will be good, and if it is wrong, it will not be possible to fit a straight line to the data. Then you have to start over with a new guess and repeat the procedure again. The linearization method depends directly on the type of function you want to linearize.

If you have a power function $y = C + ax^n$, it can be linearized by moving the constant C to the left side and then taking the logarithm of both sides:

$$\ln(y - C) = \ln(a) + n\ln(x)$$

Introducing $\ln(y - C) = Y$, $\ln(x) = X$, and $\ln(a) = b$, we get:

$$Y = b + nX$$

i.e., a linear relationship between Y and X. We see that the slope of the line is n and the intercept with the y-axis gives a.

The procedure is similar for an exponential function. To linearize, you can do the following:

$$y = C + a^x$$

Move the constant and take the logarithm:

$$\ln(y - C) = x\ln(a)$$

Introducing $\ln(y - C) = Y$ and $\ln(a) = b$, we get:

$$Y = bx$$

i.e., a linear relationship between Y and x. We see that the line passes through the origin and has the slope b.

The third and last type of function we discuss is composed of two power functions, $y = ax^n + bx^m$. A method to linearize the function is illustrated by the following example.

Example 2. In an experiment, the following values of the quantities x and y were measured:

We start by plotting an overview curve, i.e., we plot y as a function of x.

It is clear that the measurement series cannot be described by the function $y = ax^n$. It is reasonable to assume that the function can be described by $y = ax^n + bx^m$, where the function value is dominated by the first term, ax^n, for small x and by the second term, bx^m, for large x. For small x, $y \approx ax^n + 0$, and for large x, $y \approx 0 + bx^m$.

The table is supplemented with rows for $\ln(x)$ and $\ln(y)$.

We plot $\ln(y)$ as a function of $\ln(x)$ and see if we can find something we recognize.

From the graph, we see that we get two straight lines, one for small x and one for large x. With the help of these lines, the exponents n and m can be

Table 2.3: Value table.

x	y
0.1	40
0.2	21
0.4	11
0.8	6.5
1.0	6.0
2.0	6.0
4.0	9.0
6.0	13
8.0	16.4
10.0	20.4

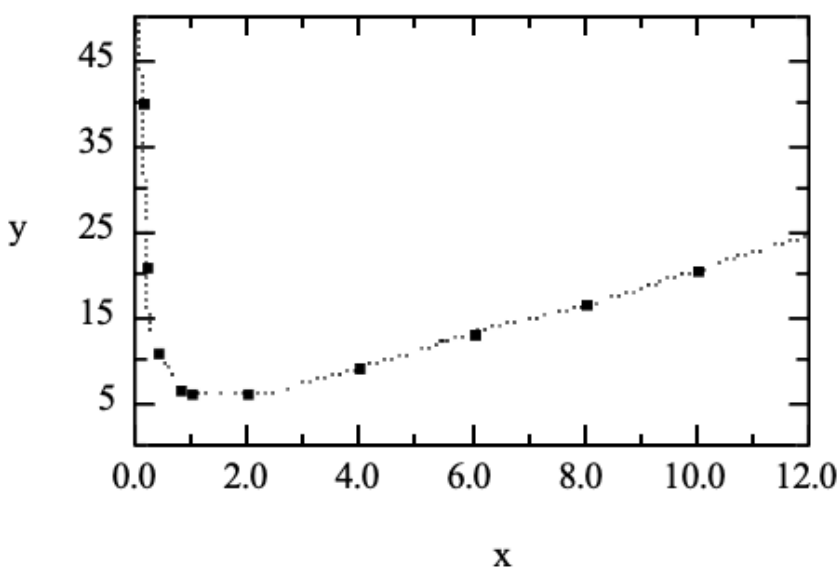


Figure 2.6: Diagram of data in Example 2.

determined. We plot each part of the graph separately in its own diagram and only include the points that belong to the straight-line fit.

For small x, we get from the graph above that $n =$ the slope of the line ≈ -1 (the logarithm gives $\ln(y) \approx \ln(a) + n\ln(x)$). From the graph for large x, we get $m =$ the slope of the line ≈ 1. The functional relationship then

Table 2.4: Value table.

x	y	$\ln(x)$	$\ln(y)$
0.1	40	-2.303	3.689
0.2	21	-1.609	3.045
0.4	11	-0.916	2.398
0.8	6.5	-0.223	1.872
1.0	6.0	0.000	1.792
2.0	6.0	0.693	1.792
4.0	9.0	1.386	2.197
6.0	13	1.792	2.565
8.0	16.4	2.079	2.797
10.0	20.4	2.303	3.016

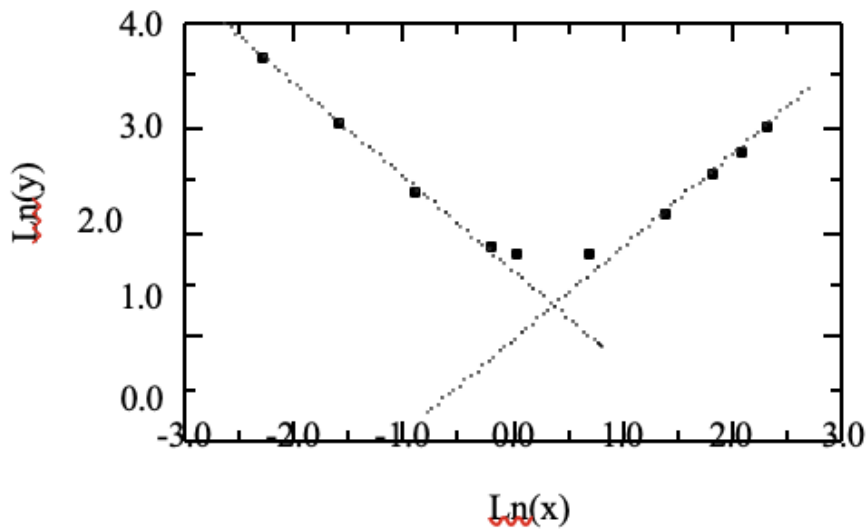


Figure 2.7: Diagram of logarithmic data. The lines (dashed) are linear curve fits.

becomes $y = ax^{-1} + bx = \frac{a}{x} + bx$.

The equations for each line are found in the diagram. For small x, from the graph above, we get $n =$ the slope of the line ≈ -1 (since logarithmically, $\ln(y) \approx \ln(a) + n\ln(x)$). For large x, we get $m =$ the slope of the line ≈ 1. The functional relationship then becomes $y = ax^{-1} + bx = a/x + bx$. To

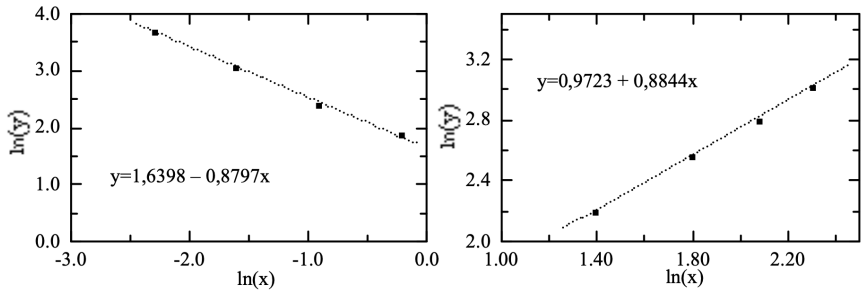


Figure 2.8: Diagram of the curve fits in Figure 2.6. The equations for the respective lines are written in the diagrams.

see if this function also describes the points that do not lay on the straight lines and to determine the constants a and b, the function is linearized. The relationship $y = a/x + bx$ can be written as $yx = a + bx^2$. Introducing the variables $yx = Y$ and $x^2 = X$, we get the equation for a straight line $Y = bX + a$, where b is the slope and a is obtained for $X = 0$. Thus, we have linearized our original equation. We need to complement our value table for x and y with values for yx and x^2 and plot these values in a coordinate system.

Table 2.5: Value table.

x	x^2	yx	y
0.1	0.01	4.0	40
0.2	0.04	4.2	21
0.4	0.16	4.4	11
0.8	0.64	5.2	6.5
1.0	1.0	6.0	6.0
2.0	4.0	12.0	6.0
4.0	16.0	36.0	9.0
6.0	36.0	78.0	13
8.0	64.0	131.2	16.4
10.0	100.0	204	20.4

Reading from the diagram gives $a \approx 4$ and $b \approx 2$. The sought function becomes $y = 4/x + 2x$. When linearizing, scales can be chosen freely. Note that the choice of scales affects your view of the actual conditions. Some

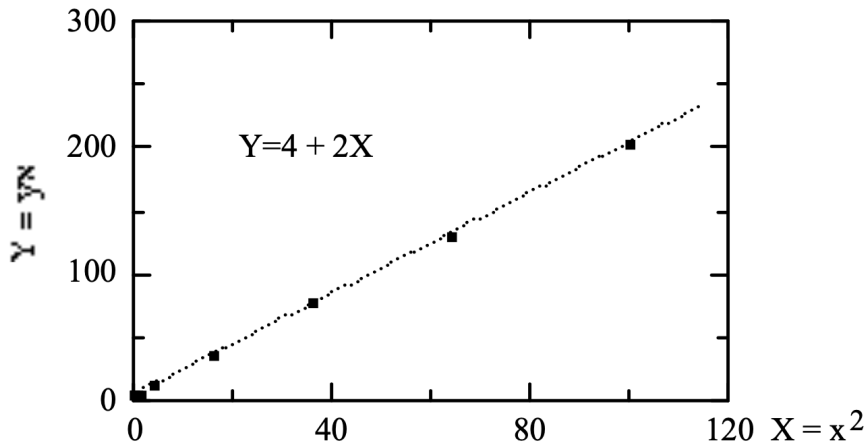


Figure 2.9: Diagram of Y as a function of X. The diagram shows that intermediate points are well fitted to the function.

scales can give a completely distorted picture of reality.

2.4 A Special Technique

A special technique used when there are not enough measurement values is illustrated in the following example.

Example 3: To determine how the period of a homogeneous flywheel depends on the wheel's properties, a string is hooked onto a screw on the wheel's periphery, wound a few turns around the wheel, and a weight with mass m is attached to the other end. When the weight is released, it moves downward and causes the wheel to rotate. After the weight has moved a certain distance s downward, the wheel's period is measured. How does the period t depend on the wheel's properties, i.e., its radius R, thickness h, and density ρ? We have 11 flywheels with the following values for R, h, and ρ.

To find a relationship between several different variables, it is appropriate to assume that the relationship is of the type $t = CR^{\alpha}h^{E}\rho^{J}$, where C is a constant. To determine any of the exponents α, E, or J, it is necessary to find measurement results where only one of the independent variables changes and the others are constant. For wheels 1–4, h and ρ are constant, and only R varies. We can then write $t = C_1R^{\alpha}$, where the new constant

Table 2.6: Wheel data.

Wheel No.	$R \times 10^{-2}$m	$h \times 10^{-2}$m	$\rho \times 10^3$kg/m^3
1	4.30	2.91	8.80
2	6.00	2.91	8.80
3	8.00	2.91	8.80
4	11.50	2.91	8.80
5	5.50	1.95	8.80
6	7.50	2.30	8.80
7	10.00	0.50	8.80
8	10.00	1.25	8.80
9	12.20	3.00	2.70
10	9.00	2.50	7.14
11	6.80	2.00	19.3

$C_1 = Ch^E\rho^J$ since both h and ρ are constant. With these measurements, we can find a relationship between t and R.

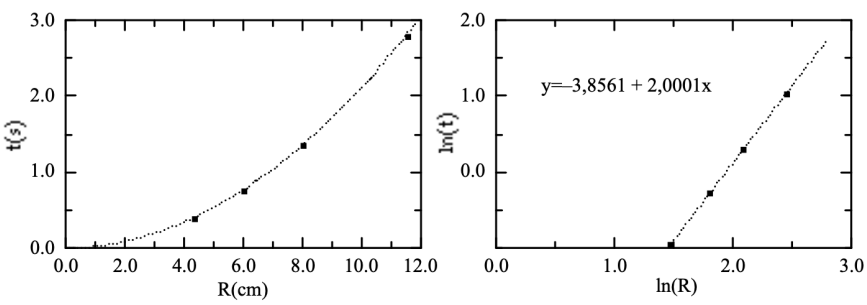


Figure 2.10: Left: wheels 1–4, time as a function of radius. Right: wheels 1–4, linearized data. The dashed line's equation is written in the diagram.

The overview curve shows that our assumption is reasonable and that it appears to be a power function with a positive exponent. Linearization (see the right curve in Figure 2.10) shows that $\alpha = 2$. The constant C_1 depends on the chosen values of h and ρ and is not of interest.

If the density ρ is constant, the assumed relationship becomes $t = C_2R^2h^E$, where $C_2 = C\rho^J$ and C_2 is constant. We do not have enough measurement values in the table to keep the radius constant. We need at least four measurement points to determine the exponent E. Since the dependence on the

Table 2.7: Measured periods for different wheels when $m = 0.020$kg and $s = 0.905$m.

Wheel No.	t(s)
1	0.391
2	0.761
3	1.356
4	2.795
5	0.523
6	1.057
7	0.876
8	1.385
9	1.769
10	1.429
11	1.120

radius is known, we can rewrite the assumed relationship as $t/R^2 = C_2 h^E$ and let the radius vary, using t/R^2 as a variable. We can then use the measurements of wheels 4-8 to determine E. Note that $t \to 0$ when $h \to 0$. The overview graph below confirms the assumption.

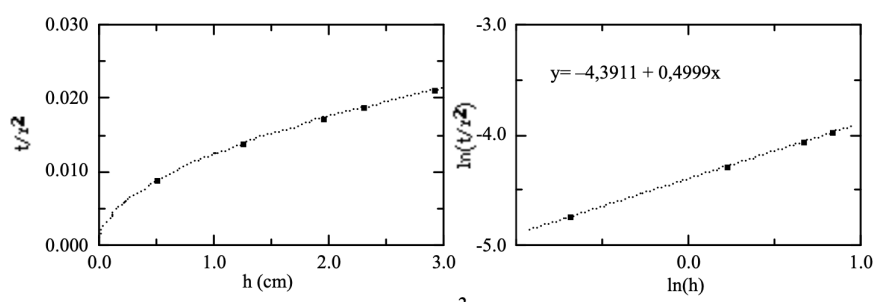


Figure 2.11: Left: wheels 4–8, t/R^2 as a function of height h. Right: linearized data, the dashed line's equation is in the diagram.

Logarithmization gives the relationship $\ln(t/R^2) = \ln(C_2) + E\ln(h)$, and the right diagram above gives $E = 0.5$. All clear! The relationship is thus $t/R^2 = C_2 h^{0.5}$, i.e., $t = C_2 R^2 h^{0.5}$. The value of the constant C_2 depends on the density and must be determined later.

2.5 Exercises

In all exercises, it is assumed that the exponent in a power expression only takes integer, half-integer, or fractional values.

Exercise 2.1

Find the relationship between the measured variables x and y. Then verify the relationship through linearization.

a) x — 0.00 1.00 2.00 3.00 4.00 5.00
y — 0.00 2.00 2.83 3.46 4.00 4.47

b) x — 0.00 1.50 2.00 3.00 5.00
y — 1.00 1.92 2.41 3.60 6.59

Exercise 2.2

The following variable values were measured in an experiment:

x	0.20	0.70	1.25	1.50	1.70
y	2.03	3.37	9.84	15.52	21.70

Furthermore, it can be assumed that $y \to 2$ when $x \to 0$. Determine the relationship between x and y and indicate how it should be transformed to be represented by a straight line.

Exercise 2.3

For the planets' average distance to the sun and their orbital periods, the following measured values are found in the almanac:

Show that the orbital period is a power function of the average distance and determine this functional relationship.

Exercise 2.4

Describe the variable m as a function of the variable L given the following measured values:

Planet	Orbital Period T (years)	Average Distance from the Sun R (AU)
Mercury	0.2408	0.3871
Venus	0.6152	0.7233
Earth	1.0000	1.0000
Mars	1.8809	1.5237
Jupiter	11.8622	5.2028
Saturn	29.4577	9.540
Uranus	84.013	19.18
Neptune	164.79	30.07

L	m
0.49	53
0.57	83
0.75	190
0.82	250
0.96	400

Exercise 2.5

Linearize the following relationships. What should be plotted as a function of what for the relationship to be linear? a, b, and k are constants.

1. $y = a + bx^2$
2. $y = a + b\sqrt{x}$
3. $y = a + bx$
4. $y = ax^b$
5. $y = ax + bx^2$
6. $y = ae^{bx}$
7. $y = a + be^{kx}$

Exercise 2.6

During a laboratory experiment in thermodynamics, the cooling of an aluminum vessel with water was studied. The temperature difference between the vessel and the surroundings $T - T_0$ was noted at certain times t.

Time t (min)	Temperature Difference $T - T_0$ (°C)
0	63
4.0	57
12.0	48
17.0	44
27.0	36

Fit the exponential function $T - T_0 = Ae^{-kt}$ to the measured values through linearization.

Exercise 2.7

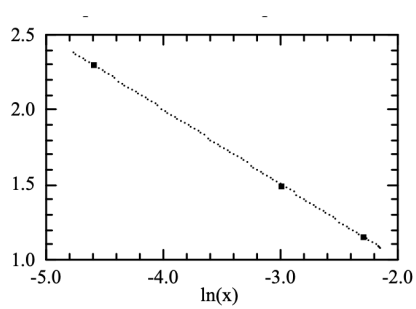
The total population of the Earth was estimated to be at least 3.5 and at most 3.6 billion in 1976, and at least 4.1 and at most 4.2 billion in 1984. Based on these values and assuming that the population grows exponentially over time, predictions for the future can be made. The results will depend on which of the given values for the two years are chosen.

1. Determine, using linearization, the lowest and highest values for the Earth's population according to this model.

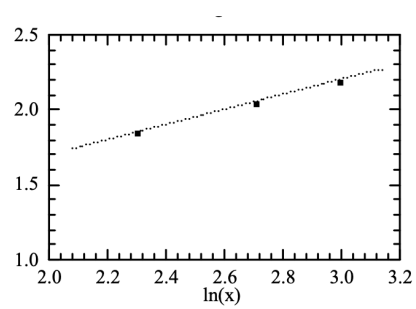
2. Indicate when the population will reach 6 billion at the earliest and latest.

Exercise 2.8

For small and large values of the variable x, the following linearizations were obtained:



Left: $\ln(y)$ as a function of $\ln(x)$ for small x.



Right: $\ln(y)$ as a function of $\ln(x)$ for large x.

1. The relationship between y and x is of the type $y = ax^n$. Determine n for small and large x.

2. The measurement series has the following values for the variables x and y:

x	y
0.01	10.2
0.05	4.92
0.10	3.79
0.20	3.13
0.50	2.83
1.0	3.00
2.0	3.54
3.0	4.04
5.0	4.92
7.0	5.67
10.0	6.67
15.0	8.00
20.0	9.17

   In exercise (1.), only the first three and the last three measured values were used. It is likely that the function $y = g(x)$ that describes the entire measurement series is a sum of two terms where one dominates for small x and the other for large x. Two different models for the function $y = g(x)$ can be considered. Which ones?

3. Determine through linearization which of the models applies to this measurement series and specify the functional relationship.

Exercise 2.9

When measuring the variables x and y, the following values were obtained:

x	y
0.1	100.5
0.2	51.0
0.3	34.8
0.5	22.5
1.0	15.0
2.0	15.0
5.0	27.0
10.0	51.0
15.0	75.7
20.0	100.5

It is reasonable to assume that the functional relationship $y = g(x)$ is such that there are two terms, one of which dominates for small x while the other dominates for large x. Determine the functional relationship.

Exercise 2.10

A weight with mass m is attached to one end of a string. The other end is hooked onto a screw on the periphery of a homogeneous flywheel, after which the string is wound around the wheel. The weight is then released, and when the weight has moved a certain distance s downwards, the flywheel's rotation period is measured. When (s) is 0.905 m, the following values for the rotation period (t) and mass (m) are obtained:

m ($10^{-3}kg$))	10.0	15.0	20.0	25.0	30.0
t (s)	1.927	1.567	1.356	1.213	1.108

When the mass (m) is 0.020 kg, the following values for time (t) and distance (s) are obtained:

s (m)	0.315	0.565	0.705	0.905	1.190	1.500
t (s)	2.304	1.716	1.541	1.356	1.181	1.054

How does the time (t) depend on (m) and (s)? Provide a single relationship that indicates (t)'s dependence on both (m) and (s) simultaneously.

2.5.1 Answers

a) $y = 2\sqrt{x}$

b) $y = 1 + 0.5x^{1.5}$

2.2 $y = 2 + 4x^3$

2.3 $t = 1.00r^{1.5}$

2.4 $m = 453L^3$

2.5
a) $y = g(x^2)$
b) $y = g(\sqrt{x})$
c) $y^2 = g(x)$
d) $y\sqrt{x} = g(x)$
e) $y^2x = g(x^2)$
f) $\ln y = g(x)$
g) $\ln(y - a) = g(x)$

2.6 $T - T_0 = 63e^{-0.021t}$

2.7
a) min = 5.9 billions, max = 6.9 billions
b) 1994 resp 2002

2.8
a) -0.5 resp. 0.5
b) $y = \sqrt{a/x + bx}$ resp. $y = \dfrac{a}{\sqrt{x}} + b\sqrt{x}$
c) $y = \dfrac{1}{\sqrt{x}} + 2\sqrt{x}$

2.9 $y = 10/x + 5x$

2.10 $t = \dfrac{0.18}{\sqrt{ms}}$

Chapter 3

Quantity, Measurement, Unit, Dimension

3.1 Introduction

A quantity is an expression of a property of an object or phenomenon that can be measured or calculated. The quantity is determined using a reference value (a unit) and a measurement value:

$$\text{QUANTITY} = \text{MEASUREMENT VALUE} \times \text{UNIT} \tag{3.1}$$

Example 1: A weight has a mass of 5 kilograms (kg). This is expressed as:

$$m = 5\,\text{kg} \tag{3.2}$$

where m is the quantity symbol, 5 is the measurement value, and kg is the unit symbol.

If the measurement value is unknown, it is written as $m = x\,\text{kg}$, where x is the measurement value symbol. Different symbols are used for a quantity and its measurement value. A physical equation can be written as a relationship between quantities.

Example 2: Newton's law of gravitation gives the force F on Earth with mass m from the Sun with mass M. The distance between Earth and the Sun is r. The relationship is:

$$F = k\frac{mM}{r^2} \tag{3.3}$$

where F, m, M, and r are quantities and k is a constant.

Example 3: Newton's second law gives the relationship between the force F on a body with mass m and the body's acceleration a:

$$F = kma \tag{3.4}$$

where F, m, and a are quantities and k is a constant.

For the result of a measurement to be relevant, it is important that the unit used is reliable because the measurement value depends on the reference. To obtain a practical and universal system, fundamental units have been introduced for a few quantities. Other units can be defined based on the fundamental units. The fundamental units must be carefully defined, internationally usable, and related to the decimal system.

In Sweden, a measurement unit system called Système International d'Unités, abbreviated as SI, is used. SI is a system established as a national standard in most countries. SI is based on older metric systems and has been designed to be used in all contexts, scientific, technical, and everyday.

3.2 SI System's Fundamental Units and Supplementary Units

The SI system is based on units for seven different quantities:

Table 3.1: SI system's 7 fundamental units.

Quantity	SI Unit	Symbol
Length	meter	m
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Luminous intensity	candela	cd
Amount of substance	mole	mol
Mass	kilogram	kg

Additionally, there are two supplementary units, radian and steradian, for the quantities plane angle and solid angle, respectively.

Plane angle: 1 radian (symbol rad) is the (plane) angle for the circular sector where the arc length is equal to the radius.

Solid angle: 1 steradian (symbol sr) is the solid angle of a cone that has its apex at the center of a sphere and cuts out an area on the sphere's surface equal to the area of a square with a side equal to the sphere's radius.

As a result of these definitions, the circumference of a circle with radius a is $2\pi a$ and the area of a sphere with radius a is $4\pi a^2$.

3.3 Derived Units

To define a derived unit, one starts from a relationship where the corresponding quantity is involved. The units for all other quantities must be established, either as fundamental units or as previously derived units.

Example 4: The unit for area.

The area A of a square with side a is given by the relationship:

$$A = ka^2 \tag{3.5}$$

By choosing the constant k to be 1, the unit for area becomes 'square meter' (m^2). 1 m^2 is equal to the area of a square with a side of 1 m.

Example 5: Velocity.

Velocity v is defined as the change in length Δs per unit time Δt, i.e.,

$$v = \frac{\Delta s}{\Delta t} \tag{3.6}$$

Starting from the units for length and time, the unit for velocity is m/s, or $\mathrm{m} \cdot \mathrm{s}^{-1}$, which is read as meters per second.

Example 6: Force.

In the SI system, Newton's second law is used to define the unit for force, named Newton. By setting the constant k to 1 in the relationship $F = kma$, we get:

$$\text{Newton} = (\text{kilogram})(\text{meter per second squared}), \text{or with symbols:} \tag{3.7}$$

$$\mathrm{N} = \mathrm{kg} \cdot \mathrm{m/s}^2 \tag{3.8}$$

Note that the constant k in Newton's law of gravitation is not set to 1; Newton's gravitational constant is 6.659×10^{-11} kg^{-1} m^3 s^{-2}. It is said that the unit N has been calibrated using Newton's second law, where the value of the constant has been established. The constant in the law of gravitation is only approximately known, as it is not fixed.

Summary: A derived unit is formed as a power or as a product of powers of fundamental units and supplementary units according to certain established physical laws for relationships between different quantities.

Nineteen derived SI units have currently been given their own names.

3.4 Multiple Units

To express a measurement value that is large or small relative to the fundamental unit or derived unit, positive or negative powers of ten can be used. Using so-called multiple units makes this significantly easier. The numerical factor is then expressed with a prefix indicating by which power of ten the unit should be multiplied.

Table 3.2: Multiple units and prefixes.

Prefix	Numerical Factor	Symbol
exa	10^{18}	E
peta	10^{15}	P
tera	10^{12}	T
giga	10^{9}	G
mega	10^{6}	M
kilo	10^{3}	k
hekto	10^{2}	h
deka	10^{1}	da
deci	10^{-1}	d
centi	10^{-2}	c
milli	10^{-3}	m
mikro	10^{-6}	μ
nano	10^{-9}	n
piko	10^{-12}	p
femto	10^{-15}	f
atto	10^{-18}	a

Most of the symbols come from Greek and Latin, but two of the units come from Nordic languages, femto (fifteen) and atto (eighteen). Prefixes representing 10 raised to a power that is a multiple of 3 should be chosen primarily. The prefixes hekto, deka, deci, and centi should therefore be avoided.

Example 7: 2,500 J = 2.5 kJ

Example 8: 0.000 001 A = 10^{-6} A = 1 μA

Example 9: The exponent for a multiple unit also applies to the prefix, i.e., 1 km^2 = 1 km · 1 km

Example 10: Prefixes can also be used for money. 200,000 kr = 200 kkr.

3.5 Dimension of Quantities

We will use the SI system when defining the concept of dimension. We assume that it is intuitively clear what dimensions the different fundamental units have; in all length measurements with different measuring sticks, a quantity with the dimension length is measured, in all time measurements with different clocks, a quantity with the dimension time is measured, and so on.

For the fundamental units in SI, the following symbols are used for the corresponding dimensions:

$$\begin{aligned}
[\text{length}] &= L \\
[\text{mass}] &= M \\
[\text{time}] &= T \\
[\text{electric current}] &= I \\
[\text{temperature}] &= \Theta \\
[\text{amount of substance}] &= N \\
[\text{luminous intensity}] &= J
\end{aligned}$$

Note that square brackets, [...], are used to indicate the dimension of the quantity within the brackets. In mechanics, only the dimensions L, M, and T appear. A mechanical quantity Q thus has a dimension that can be written as:

$$[Q] = L^{\alpha} M^{\beta} T^{\gamma} \tag{3.9}$$

A quantity Q with $\alpha = \beta = \gamma = 0$ has the dimension 1 (one) and is commonly referred to as dimensionless or dimension-free.

Example 11: The area A has $[A] = L^2$, the angle α has $[\alpha] = 1$, the velocity v has $[v] = LT^{-1}$, and the force F has $[F] = MLT^{-2}$.

3.6 Dimension Control

All physical formulas can be checked with dimensional analysis.

Example 12: A particle with mass m falls straight down a distance s and reaches the velocity v. The initial velocity is 0 and the gravitational acceleration is g.

$$v = \sqrt{2sg} \tag{3.10}$$

Check the expression for v:

$$[v] = LT^{-1}$$
$$\left[\sqrt{2sg}\right] = \left(LLT^{-2}\right)^{1/2} = LT^{-1}$$

Thus, $[v] = [\sqrt{2sg}]$ and the expression for v can be correct.

The calculation rules for the dimension of integrals and derivatives are the same as the corresponding calculation rules for products and quotients.

Example 13:

$$v = \frac{dx}{dt} \tag{3.11}$$

where x is distance and t is time, so $[v] = [x]/[t] = L/T$.

Example 14: By definition, the moment of inertia $I = \int r^2 \rho dV$, where r is distance, ρ is density, and V is volume. It follows that:

$$[I] = L^2 M L^{-3} L^3 = L^2 M \tag{3.12}$$

The unit for I is kilogram meter squared, kg $\cdot$ m^2.

Example 15: A container shaped like a horizontal cylinder is completely filled with gasoline. It has an internal length of 210 cm and a diameter of 128 cm. At the top, there is a hole for air, and the container can be drained through a circular hole with a diameter of 60 mm located at the bottom and opening straight down. Calculate the time for emptying, given that the contraction coefficient k is 0.90.

Symbols: The container's length is L, the container's diameter is D, the hole's diameter is d, the contraction coefficient is k (dimensionless quantity), the gravitational acceleration is g, the height of the liquid surface above the bottom of the container is x, the volume of liquid in the container is V, and the outflow time is τ.

$$\tau = \frac{8LD^{3/2}}{3\pi k\sqrt{gd^2}} \tag{3.13}$$

We find:

$$\begin{aligned}
[8] &= 1, \\
[L] &= L, \\
[D^{3/2}] &= L^{3/2}, \\
[3\pi] &= 1, \\
[k] &= 1, \\
[\sqrt{g}] &= L^{1/2}T^{-1}, \\
[d^2] &= L^2
\end{aligned}$$

and the right-hand side becomes:

$$[HL] = LL^{3/2}/(L^{1/2}T^{-1}L^2) = T, \tag{3.14}$$

which is equal to the dimension of the left-hand side.

3.7 Preliminary Dimensional Analysis

As mentioned earlier, all physical formulas can be checked with dimensional analysis. Dimensional analysis can also be used to obtain a dimensionally correct formula. We illustrate this with an example.

Example 16: We want a mathematical model for the orbital period of planets around the Sun. We assume the planetary orbit is elliptical and have strong reasons to believe that the orbital period (dependent variable) t depends on the length of the major axis a, the mass of the Sun m_s, and the gravitational constant G (which has the unit $\mathrm{Nm}^2/\mathrm{kg}^2$). However, we have found that the orbital period of a planet does not depend on its mass. Our variable table, where the dependent variable is placed at the top, is shown below.

Table 3.3: Variable table for Example 16.

Quantity	Symbol	Dimension
Orbital period	t	T
Major axis length	a	L
Sun's mass	m_s	M
Gravitational constant	G	$\mathrm{L}^3\mathrm{T}^{-2}\mathrm{M}^{-1}$

The orbital period can be written as a function Φ of the independent variables according to:

$$T = \Phi(a, m_s, G) \tag{3.15}$$

We guess, with good reason, that we can make a product assumption (factorial power function):

$$T = C \cdot a^{\alpha} \cdot m_s^{\beta} \cdot G^{\gamma}, \tag{3.16}$$

where C is a dimensionless constant and α, β, and γ are exponents.

It should hold that $[VL] = [HL]$. This means that we can determine as many exponents as we have fundamental dimensions in the assumption.

$[VL] = [HL]$ gives:

$$T = L^{\alpha} \cdot M^{\beta} \cdot (L^3 T^{-2} M^{-1})^{\gamma} \tag{3.17}$$

Thus:

$$\begin{aligned} T: &\quad 1 = -2\gamma \quad \Rightarrow \quad \gamma = -\frac{1}{2} \\ L: &\quad 0 = \alpha + 3\gamma \quad \Rightarrow \quad \alpha = -\frac{3}{2} \\ M: &\quad 0 = \beta - \gamma \quad \Rightarrow \quad \beta = \gamma = -\frac{1}{2} \end{aligned}$$

A dimensionally correct model according to our assumption is then:

$$T = C \frac{a^{3/2}}{\sqrt{G m_s}} \tag{3.18}$$

There will be more on dimensional analysis in Physics 2 (Chapter 5).

3.8 Exercises

1. String Force. A particle with mass m is attached to a string and moves in a circular path on a smooth table. The particle's speed is v and the radius of the path is r. The radius is constant. Derive an expression for the force F with which the string affects the particle.

2. Write out the following multiple units in numbers based on the filled example:

- 1 kilowatt (1 kW) = 1000 W

- 1 micrometer (1 μm) = ... m
- 1 megajoule (1 MJ) = ... J
- 1 milligram (1 mg) = ... g

3. Rewrite the measurement data following the example for the first two cases:

- 50 km = 50 $\cdot 10^3$ m
- $2 \cdot 10^7$ Pa = 20 MPa
- 180 MPa = ... Pa
- 5 TJ = ... J
- 50 μm = ... m
- 1 nm = ...m
- 50 kW = ...W
- 350 cm^3 = ...m^3
- 220 GPa = ...Pa
- 2000 N = ... kN
- 0.00003 m = ... μm
- 10^{11} Pa = ...GPa
- 10^{13} J = ...TJ

4. How are the following additional units defined and in what contexts are they used?

- ångström
- parsek
- barn
- universal mass unit

- electronvolt
- gauss

5. Perform the following unit conversions:

- 1 N = ... dyn
- 1 hk = ... W
- 1 atm = ... torr
- 1 kWh = ... J
- 1 kp = ... N
- 1 kcal/h = ... W
- 1 eV = ... J
- 100°C = ... K
- 1 kcal = ... J

6. In solving a problem, the following final expression for the shear modulus was obtained with the following notations:

$$G = \frac{16\pi sm(d_1^2 + d_2^2)}{d^4(T_2^2 - T_1^2)} \tag{3.19}$$

Notations: Wire length s, wire diameter d, wire shear modulus G, mass of the applied ring m, oscillation period without ring T_1, oscillation period with ring T_2, ring outer diameter d_1, and ring inner diameter d_2. Determine through dimensional analysis if the final expression can be correct.

3.9 Answers to Exercises

Exercise Answer

- 1 $F = C\frac{mv^2}{r}$, where C is a dimensionless constant.
- 2-6 -

Chapter 4

Elementary Mechanics and Fluid Dynamics

4.1 Strain of Solid Bodies

All bodies deform under forces. A body that returns to its original shape when the forces are removed is said to behave elastically. Consider a rod with length s and cross-sectional area A. The rod is fixed at one end. When the rod is subjected to an axial tensile force F, it elongates by a distance Δs.

Internal forces arise in the rod. To describe the internal forces, one can imagine the rod being divided by a cut perpendicular to its length. On the cut-off part, forces corresponding to a normal stress σ act from the other part of the rod. The cut-off part of the rod is in equilibrium.

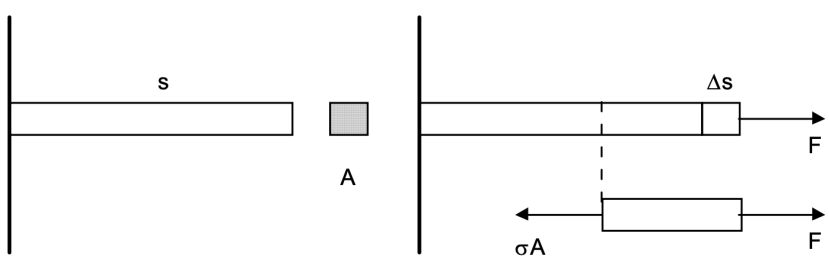


Figure 4.1: Axial strain of an elastic rod.

The sum of the forces is therefore zero, i.e.,

$$F - \sigma A = 0 \tag{4.1}$$

During deformation, each unit length of the rod elongates equally. The strain ϵ is defined as the relative length change:

$$\epsilon = \frac{\Delta s}{s} \tag{4.2}$$

The strain is thus dimensionless, $[\epsilon] = 1$.

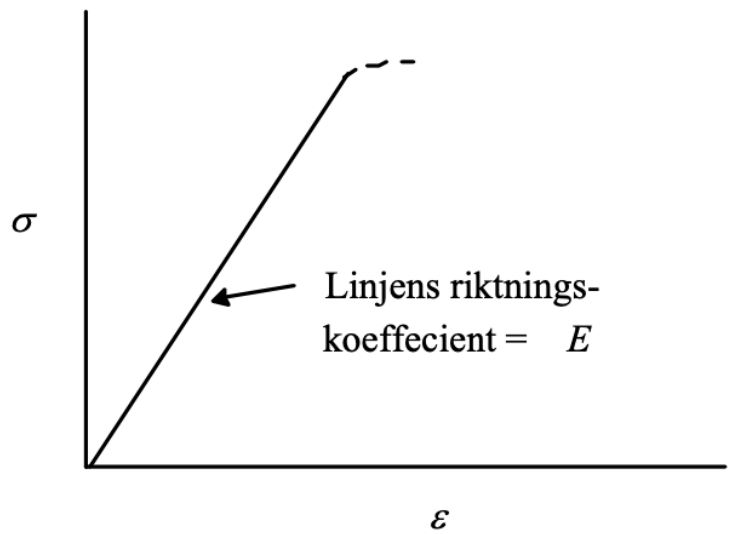


Figure 4.2: Stress-strain diagram for a linear elastic material.

As long as the deformation is elastic, the stress depends only on the strain. For many materials, it holds that σ is proportional to ϵ within a large part of the elastic region, i.e.,

$$\sigma = E\epsilon \quad \text{(Hooke's law)} \tag{4.3}$$

The proportionality constant E is called the modulus of elasticity. It is a material constant that is tabulated for metals and other solid substances. Since ϵ is dimensionless, E has the same dimension as σ,

$$[E] = ML^{-1}T^{-2} \tag{4.4}$$

E is usually given in the unit N/m^2 (= Pascal, Pa). As shown by the definitions above, Hooke's law can also be written in the dimensionless form:

$$\frac{\Delta s}{s} = \frac{F}{EA} \tag{4.5}$$

4.2 Shear of Solid Bodies

A rectangular block is fixed to a flat surface. When the upper surface with area A is subjected to a force F, parallel to one edge, the block deforms as shown in the figure below. This type of deformation is called shear. During shear, the layers tend to slide relative to each other. In an imaginary cut through a layer parallel to the upper surface with area A, a tangential force distribution given by the shear stress $\tau = F/A$ arises. The deformation is indicated by the shear $\gamma = \Delta s/s = \tan \Delta\theta$.

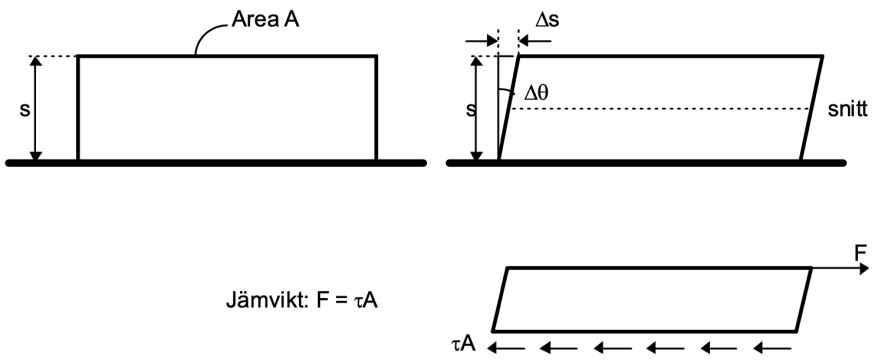


Figure 4.3: Shear of an elastic block. The block is sheared by the force F and the tangential shear stress τ is opposite to the force.

Under moderate loads, the material usually behaves linearly elastically, which means that

$$\tau = G\gamma = G\Delta\theta \quad \text{or} \quad \Delta\theta = \frac{F}{GA} \tag{4.6}$$

The proportionality constant G, the shear modulus (or torsion modulus), is a material constant with the dimension of force per unit area, i.e.,

$$[G] = ML^{-1}T^{-2} \tag{4.7}$$

Shear also occurs during the twisting of wires and shafts. The figure below shows a wire twisted under the influence of the torque M. For a wire with a circular cross-section, it holds that

$$\Delta\theta = \frac{2Ms}{G\pi r^4} \tag{4.8}$$

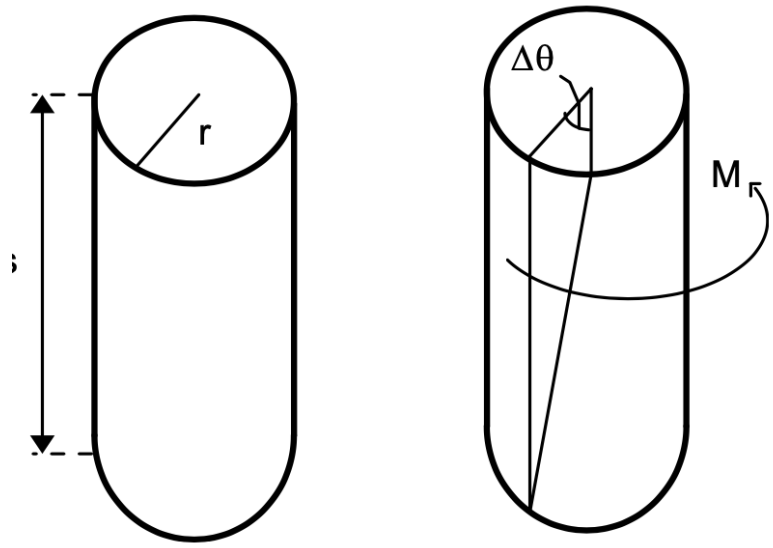


Figure 4.4: The rod is subjected to a torque and deforms, shearing.

4.3 Relationship Between E and G

When a rod is stretched due to an axial force F, it also narrows. Let the x-axis be directed along the rod, and the y- and z-axes be perpendicular to the rod's length. It holds that

$$\begin{aligned}\epsilon_x &= \frac{\sigma_x}{E},\\ \epsilon_y &= -\nu\frac{\sigma_x}{E},\\ \epsilon_z &= -\nu\frac{\sigma_x}{E},\end{aligned}$$

where ν is Poisson's ratio. Poisson's ratio is a constant that indicates how much the rod shrinks in the transverse direction. For many materials, Poisson's ratio is approximately 0.25. Since the force F causes both stretching and shearing, it is understood that there should be a relationship between E, G, and ν. It can be shown that

$$G = \frac{E}{2(1+\nu)} \tag{4.9}$$

4.4 Viscosity (Shearing of Liquids)

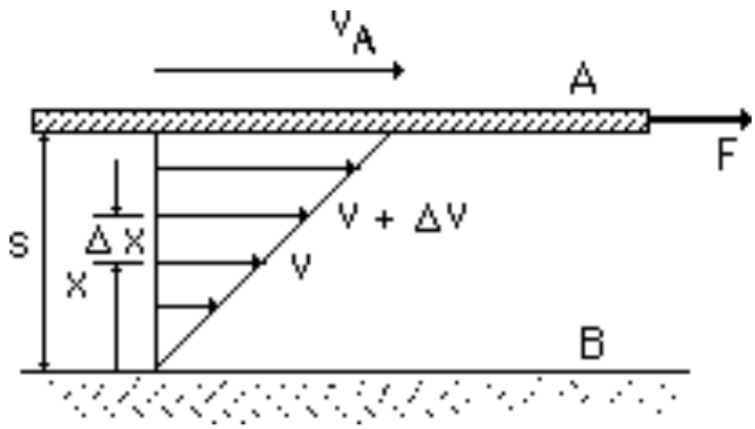


Figure 4.5: Velocity profile for time-independent, laminar Couette flow.

Let us study a liquid between two parallel plates. Plate A moves with a constant velocity v_A under the influence of the force F while plate B is stationary. The molecules of the liquid "stick" to the surfaces of the plates, as do gas and liquid molecules in general, and follow the movement of the plates. Since the liquid is viscous, having a certain "thickness," the liquid in the space between the plates will be dragged along as the upper plate moves. We assume that the flow that arises in the liquid is laminar, i.e., the liquid layers slide against each other without mixing. (Laminar flow occurs at low flow velocities. At higher velocities, the layers break up, and turbulent flow arises.)

During the flow, due to viscosity, tangential friction forces and thus shear stresses arise between the layers. The constant force F acting on plate A balances the friction forces. Equilibrium prevails, so the shear stress $\tau = F/A$. Experimentally, it is found that F/A is proportional to the plate's velocity v_A and inversely proportional to the distance s between the plates, i.e.,

$$F = \text{const.} \cdot v_A \quad \text{or} \quad \tau = \mu \frac{v_A}{s}, \tag{4.10}$$

where the proportionality constant μ is called the dynamic viscosity, $[\mu] = \mathrm{ML^{-1}T^{-1}}$ or $\mathrm{Ns/m^2}$. The resulting velocity distribution in the liquid is illustrated in the figure above. The flow velocity increases linearly with the

coordinate x. From $v = 0$ at the surface of plate B to $v = v_A$ at the surface of plate A, we get

$$\frac{v_A}{s} = \frac{\Delta v}{\Delta x} \quad \text{or after taking the limit} \quad \frac{dv}{dx}. \tag{4.11}$$

The derivative dv/dx is called the velocity gradient and indicates the rate of change of velocity per unit length perpendicular to the flow. Thus, the more general relationship for laminar flow is obtained:

$$\tau = \mu \frac{dv}{dx}. \tag{4.12}$$

The dimension of μ is determined by the above relationship; remember that $[\tau] = \mathrm{ML^{-1}T^{-2}}$ and $[dv/dx] = \mathrm{T^{-1}}$, so

$$[\mu] = \mathrm{ML^{-1}T^{-1}}. \tag{4.13}$$

In various contexts, the dynamic viscosity is used instead of the kinematic viscosity. The dynamic viscosity is defined as:

$$\nu = \frac{\mu}{\rho}, \quad [\nu] = \mathrm{L^2T^{-1}}. \tag{4.14}$$

These relationships between viscosity, shear stress, and velocity change per unit length were proposed in the mid-1600s by Isaac Newton. The liquids that obey this relationship are therefore called Newtonian fluids. Most liquids are Newtonian; exceptions include some oils and polymeric liquids.

4.5 Exercises

1. A round steel rod with a diameter of 10 mm is subjected to a tensile force of 18,000 N. What is the stress in the rod?

2. On the rod, a measurement length of 100 mm is marked. How much does the measurement length increase? The modulus of elasticity for the rod is 210,000 N/mm^2.

3. A spin fisherman accidentally gets a snag. The nylon line has a diameter of 0.70 mm and the extended length is 16 m. During the attempt to free the snag, the fisherman pulls with a force of 250 N, causing the line to elongate by 340 mm. Determine the modulus of elasticity for the nylon material in the line.

4. A rod is composed of two parts with circular cross-sections. The thinner part is made of steel and has a diameter of 10 mm, while the thicker part is made of aluminum and has a diameter of 20 mm. Both parts have a length of 200 mm. How much does the rod elongate when subjected to a force of 20,000 N? Steel has a modulus of elasticity of 200,000 N/mm^2 and aluminum 75,000 N/mm^2.

5. The figure shows a rubber block with dimensions 40 mm $\times$ 40 mm $\times$ 25 mm, which is glued between two metal plates. Under the influence of the force F, the upper plate is displaced 3.0 mm sideways. Determine the shear modulus for the rubber material if F is 200 N.

6. A small spherical raindrop has a radius of 0.05 mm. Determine the velocity of the raindrop during free fall in the air. The drop falls at a constant velocity. A small sphere with radius r moving at a low velocity v in a viscous liquid or gas is affected by this with a resistance $F = 6\pi r\mu v$ where μ is the dynamic viscosity of the surrounding medium. The relationship is called Stokes' law and applies if the Reynolds number (Re $= 2rv\rho/\mu$) is much less than one. Assume that the drop's velocity is such that Stokes' law applies.

4.6 Answers

a) 230×10^6 N/m^2

b) 0.11 mm

1. 30.6 $\times 10^9$ N/m^2

2. 0.42 mm

3. 1.04 $\times 10^6$ N/m^2

4. 0.32 m/s

Chapter 5

Dimensional Analysis

5.1 Dimensional Considerations

This section aims to familiarize you with the concept of dimensions in physics. Several examples illustrate how natural phenomena can often be described through simple dimensional considerations. Many examples are drawn from a classic in physics literature: d'Arcy Thompson's "On Growth and Form," first published in 1917.

Length, Area, and Volume

What physical effect compels a diner to cut a hot potato into pieces? Cutting a potato into pieces does not change the volume of the potato and thus the total amount of heat that must be conducted away to make it edible. However, the surface area of the potato increases, and thus the amount of heat conducted per unit time through the surface increases, resulting in faster cooling.

The same phenomenon influences the food requirements of mammals. A significant portion of the food is required to maintain body temperature. The volume of an animal (with dimension L^3) grows in proportion to the cube of its size (with dimension L). Since the body surface area depends on the square of the length (L^2), smaller animals have proportionally larger body surfaces relative to their volume and thus proportionally greater heat loss to compensate. A mouse consumes food equivalent to about half its body weight per day, while a human needs only about one-fiftieth of their body weight per day. The approximate energy turnover per kilogram of body

weight and per day in the metabolism of some mammals is:

Table 5.1: Energy requirements and weight for various animals.

Being	**Weight (kg)**	**Energy Requirement (J/(day kg))**
Guinea Pig	0.7	940
Human	70	140
Horse	600	93
Elephant	4000	54
Whale	150000	7

It is about 20 times more energy-expensive to maintain a kilogram of human than a kilogram of whale. A mammal with a mass much smaller than a dormouse would not have time to eat, digest, and metabolize the food needed to sustain its energy turnover.

In Jonathan Swift's book "Gulliver's Travels," it is told what happens when Gulliver arrives in the land of "Lilliput." When the ministers in "Lilliput" find that Gulliver is about twelve times larger than the "average Liliputian," they decide to allocate Gulliver a daily food ration proportional to his body volume, i.e., $12^3 = 1728$ normal rations in "Lilliput." Gulliver would have had considerable difficulty if he had tried to consume all that food, which was sufficient for about twelve giants of his size. His food requirement is proportional to his body surface area, i.e., it is $12^2 = 144$ normal portions. Jonathan Swift wrote the book as a social satire, but this was probably an unconscious irony over the political decision-makers' poor knowledge of scientific problem-solving.

Further examples of the surface-to-volume effect can be found in nature's way of constructing fruit stalks or bones. Large fruits (e.g., apples) have proportionally thicker stalks than small ones (e.g., blueberries), and in some cases, the delicacy is placed on the ground (e.g., cucumbers and pumpkins). The underlying cause is the same as for the bones of animals.

The bones of animals must bear the body weight. The weight is proportional to the volume (L^3) since the density is roughly the same for living matter. The bearing capacity of the bone is proportional to the cross-sectional area of the bone, i.e., roughly proportional to L^2. This favors small animals that can have elegant, slender bone constructions. Compare, for example, the legs of an elephant with the legs of an antelope. Even more markedly, some insects' (e.g., harvestmen) leg constructions differ from those of mammals. In the extreme case, one can imagine a leg so large that it cannot

support itself. The giants of the sagas were thus completely harmless and rather pitiable. They only needed to try to stand up to break their legs.

The Art of Jumping

It happens that writers, filled with admiration for the grasshopper's ability to jump, write that if a grasshopper were as large as a human, it would jump over church towers. This is, of course, unreasonable, and one can also note that in some respects, humans are superior to grasshoppers as high jumpers.

How high can one expect an animal with length L to jump? The jump height is determined by the muscle work performed during the jump. Muscle force is proportional to the cross-sectional area of the muscles, i.e., to L^2. The muscle force times the contraction length (proportional to L) determines the work, i.e.,

$$\begin{aligned} W &= \text{muscle work during the jump} = \\ &= (\text{muscle force}) \cdot (\text{contraction length}) = \\ &= \text{const} \cdot (L^2) \cdot (L) = \\ &= \text{const} \cdot M \end{aligned} \tag{5.1}$$

since the mass M is proportional to the volume.

Muscle work gives the body kinetic energy. The kinetic energy is converted to potential energy when the body moves upward in the gravitational field, and when the body reaches the highest point, the muscle work has been converted to potential energy via kinetic energy (we neglect energy losses due to friction in joints, air resistance, etc.). The potential energy, E_p, at the turning point is (where h is the jump height)

$$\begin{aligned} E_p &= (\text{body weight}) \cdot (\text{jump height}) = \\ &= \text{const} \cdot (\text{body mass}) \cdot (\text{jump height}) = \\ &= \text{const} \cdot Mh. \end{aligned} \tag{5.2}$$

According to the principle of energy conservation, W and E_p will be equal, which means that the jump height (h) is constant and independent of the mass.

A "Lilliputian" should therefore have been able to jump as high as Gulliver. More generally, one can say that all living beings that are similarly built in proportion to their length should be able to jump equally high.

The grasshopper or flea can, however, be constructed with proportionally much longer and more slender legs than, for example, a human or an elephant,

as the strength problems are less difficult to solve. Nature has also utilized this, and therefore the grasshopper is a better high jumper than humans. On the other hand, humans jump high with the "inside of the head," i.e., they (if well-trained) ensure that the body bit by bit goes over the bar in such a way that the center of gravity goes under or through the bar (the style is called the Fosbury-flop after the high jumper Richard (Dick) Fosbury who first used the style at the 1968 Mexico City Olympics). In this respect, humans are superior to grasshoppers.

5.2 Three Examples of Relationships Between Dimensionless Quantities

Every sensible relationship between physical quantities can be written as a relationship between dimensionless quantities. The original relationship contains more variables than the dimensionless one, which is therefore easier to visualize.

Example 1: Radioactive Decay

Consider a radioactive isotope whose initial mass is m_0 at time $t = 0$. The mass m remaining after time t depends on m_0 according to the equation

$$m = m_0 e^{-0.693t/t_0}, \tag{5.3}$$

where t_0 is the isotope's half-life. (Note that $\exp(-0.693) = 2^{-1}$.) The half-life t_0 is tabulated for almost every radioactive element. It is therefore possible to plot a graph where m is plotted against time t for the element in question. However, it is preferable to plot m/m_0 against t/t_0 as shown below. This curve applies generally to all radioactive elements. The equation can be written as:

$$M = e^{-0.693T} \quad \text{or} \quad M = \phi(T), \tag{5.4}$$

where

$$M = \frac{m}{m_0} \quad \text{and} \quad T = \frac{t}{t_0}. \tag{5.5}$$

Clearly, M and T are dimensionless variables corresponding to the quantities m and t. (If a curve were plotted of m against t, it would have the same shape as the curve in the figure below.)

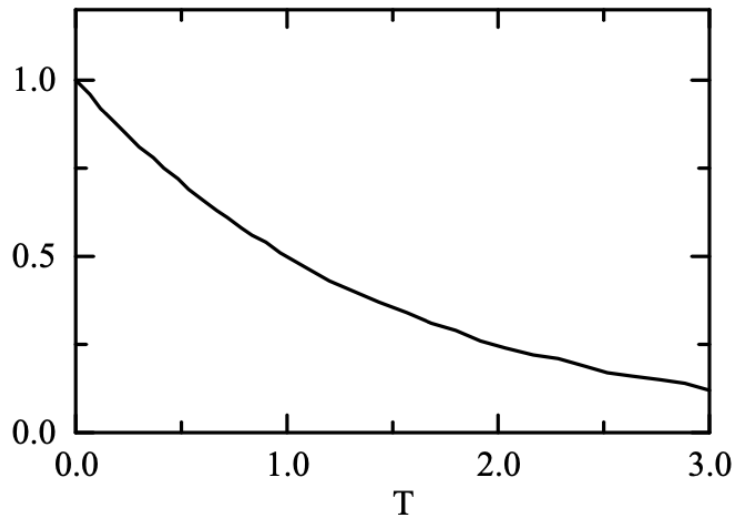


Figure 5.1: Radioactive decay in dimensionless quantities.

In this example, the function ϕ is a known function; $\phi(T) = 2^{-T}$. However, the function ϕ can be determined experimentally by measuring m as a function of t, calculating M and T, and plotting the results. The resulting curve will be the same as in Figure 5.1 above.

Example 2: The Area of a Right Circular Cone

The area of a cone's total surface consists of the base area and the lateral surface area. The total area depends on two variables, r and s, and can be written as $A = \pi r^2 + \pi rs$, or symbolically $A = \phi_1(r, s)$. The area can be represented as a family of curves:

If we divide the equation for the area A by, in the first case, πr^2 and in the second case by πrs, we get two relationships:

$$\frac{A}{\pi r^2} = 1 + \frac{s}{r} \quad \text{or} \quad \frac{A}{\pi rs} = 1 + \frac{r}{s}. \tag{5.6}$$

Graphically, this can be represented as in Figure 5.3.

The family of curves has been replaced by a single straight line that can describe all types of cones. Knowing s and r, one can calculate s/r and read $A/(\pi r^2)$ from the graph. Then A can be calculated. Note also that all cones

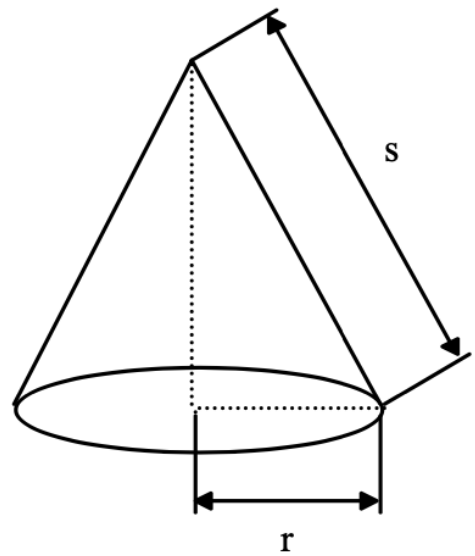


Figure 5.2: Cone.

with the same ratio s/r are represented by the same point on the straight line. For it to be a cone, s must be greater than r. Therefore, s/r varies between 1 and infinity for different cones, while r/s varies between 0 and 1. It is often advantageous to use the latter variable, which varies between 0 and 1.

Example 3: Kinematics

For motion with constant acceleration a, the distance s traveled after a time t is

$$s = vt + \frac{1}{2}at^2 \tag{5.7}$$

if the initial velocity at time $t = 0$ is v. We have four variables, and since the initial velocity is constant, this corresponds to a family of curves with three varying quantities, s, t, and a, as shown in the diagram below. For each value of the fourth variable v, one gets similar diagrams.

We will, as in the previous example, derive a corresponding dimensionless

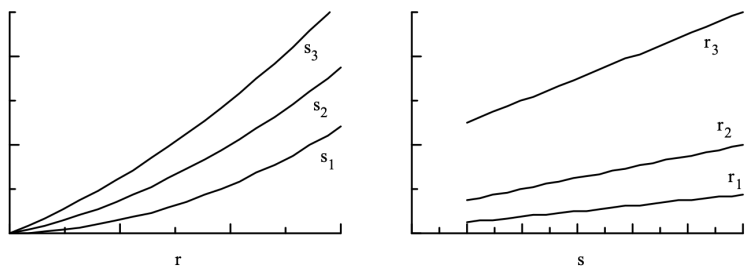


Figure 5.3: The surface area of a cone. The left diagram shows the area as a function of r with s as a parameter, and the right diagram shows the area as a function of s with r as a parameter.

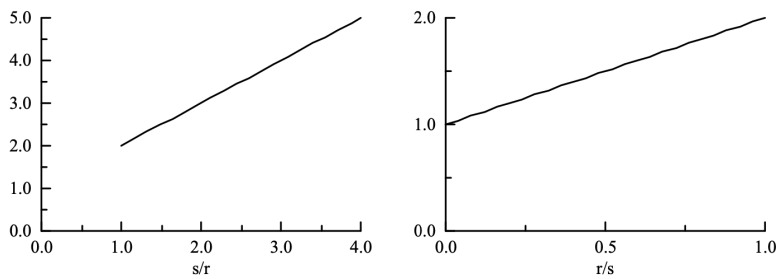


Figure 5.4: The surface area of a cone in dimensionless variables.

relationship. Divide, for example, by vt:

$$\frac{s}{vt} = 1 + \frac{at}{2v} \quad \text{or} \quad \frac{s}{vt} = \phi\left(\frac{at}{v}\right). \tag{5.8}$$

In this example, we know the function ϕ. But suppose the form of ϕ is unknown, and we need to determine s for different values of a, t, and v experimentally. The dimensionless variables $s/(vt)$ and $at/(2v)$ are calculated and plotted in a diagram. The shape of the graph now gives the function ϕ. We get the following:

The different families of curves for $v = v_1$, $v = v_2$, etc., have evidently been reduced to a single graph, which in this case happens to be a straight line.

The relationship between s, v, t, and a can be made dimensionless in different ways. Division by vt (as above) gives a form that is suitable if one wants to determine s for known a, v, and t since s is included in only one

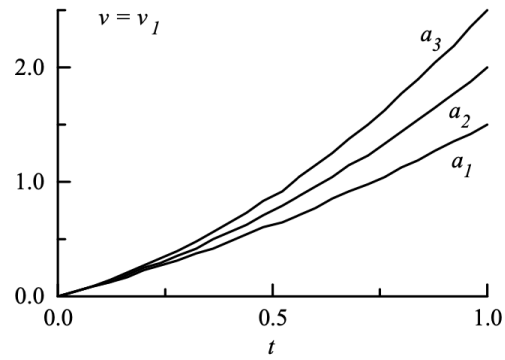


Figure 5.5: Distance traveled, s, with constant acceleration. The diagram shows distance as a function of time with acceleration as a parameter. The initial velocity is constant.

of the dimensionless variables. If instead, one divides by at^2, the equation becomes

$$\frac{s}{at^2} = \frac{v}{at} + \frac{1}{2} \quad \text{or} \quad \frac{s}{at^2} = \phi\left(\frac{v}{at}\right). \tag{5.9}$$

Here, the quantity v is included in only one of the dimensionless variables, and the corresponding graph is suitable for determining v for given s, t, and a.

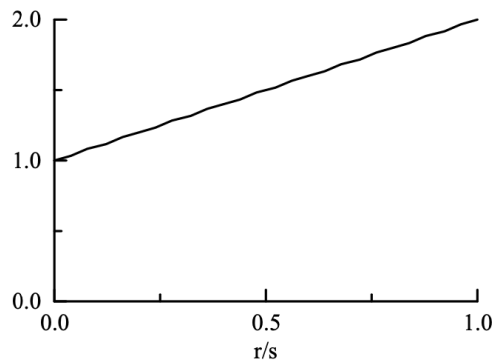


Figure 5.6: Distance traveled, s, with constant acceleration. Plotted in dimensionless variables.

Conclusions from the Examples

The examples illustrate three principles worth remembering:

1. The dimensionless relationships can be given different forms, e.g.,

$$\frac{s}{vt} = \phi\left(\frac{at}{v}\right) \quad \text{or} \quad \frac{s}{at^2} = \phi\left(\frac{v}{at}\right). \tag{5.10}$$

   Which form is suitable depends on the circumstances, e.g., whether s or v is to be calculated with the other variables known.

2. The function ϕ is determined through a special analysis or by fitting a curve to experimentally obtained points. In the examples above, the functions ϕ were known through theoretical derivations. They could have been determined through a series of experiments.

3. The number of dimensionless variables is less than the original number of variables. The reduction can be summarized in a table, see Table 5.2.

Table 5.2: Table of variables in the examples.

Example	**Number of Variables (n)**	**Number of Fundamental Dimensions (m)**	**Number of Dimensionless Variables**
"Cone"	3 (A, r, s)	1 (L)	2 = 3-1
"Kinematics"	4 (s, v, a, t)	2 (L, T)	2 = 4-2
"Decay"	4 (m, m_0, t, t_0)	2 (M, T)	2 = 4-2

5.2.1 Choice of Variables

As a rule, experiments, dimensional analysis, and ordinary mathematical analysis must be used in conjunction to create a satisfactory model. The choice of primary variables is naturally part of the modeling work. The goal is to find an equation

$$\phi(q_1, q_2, q_3, \ldots) = 0 \tag{5.11}$$

for the quantities $q_1, q_2, q_3, \ldots$, an equation that is complete and unique. Completeness means that no additional measurements are required, and uniqueness means that no additional relationship between the variables exists.

Let's provide some guidelines for choosing primary variables:

- Choose the simplest variables possible. If, say, a rod has a circular cross-section, it is better to let the diameter be the variable rather than the area.

- When an object's weight is significant, the gravitational acceleration g should be included as a variable, even though we naturally cannot vary g.

- Some quantities are dimensionless from the start, e.g., angles and ratios between lengths, masses, and times. These dimensionless variables are often suitable to choose.

- An object's shape can be characterized by a length and by ratios between lengths.

Physicists and engineers use simplifications to reduce the number of variables describing a problem. This often increases the understanding of the

problem and makes it manageable. A body is rigid, the strain is small, equilibrium prevails, the motion is time-independent, the fluid is non-viscous, the material is homogeneous, etc., are examples of approximations that make it possible to address important phenomena without getting bogged down in trivialities. The result of the simplifications is an idealized model. The use of such concepts and methods is so widespread that one often does not distinguish between the model and reality. Sometimes the result of this oversight is catastrophic. To get a simple model, it is important to reduce the number of variables and to sacrifice some accuracy. If our understanding of the problem is rudimentary, we should simplify drastically. On the other hand, if high precision is essential, as in navigating in an archipelago in fog and darkness, we must tolerate more complex and accurate models.

In general, the physical relationships being modeled can be expressed using dimensionless variables. It is often effective to combine dimensional analysis and experimental work. The advantages are:

- The number of dimensionless variables is less than the original number of variables (see Table 5.2 above).
- The effect of an original variable can be determined by varying another original variable.
- In model experiments, the scaling is evident.
- Sometimes one realizes that a supposed original variable does not affect the dependent variable, or that an essential independent variable has been forgotten.

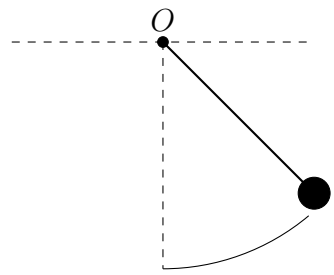
The disadvantages of the analysis are the work it entails. Sometimes it is simpler to first vary all the original variables directly, one at a time, and then check the dimensions of the final expression than to work with dimensionless variables from the start.

Obs: Skipped Sections 5.4 and 5.5 about π-groups.

5.2.2 Exercises

1. **Pendulum.** A mathematical pendulum consists of a thin string with negligible mass. The string is attached at point O. At the other end of the string, a point mass is attached, and the pendulum swings freely

in a vertical plane under the influence of gravity. Air resistance, etc., can be neglected.

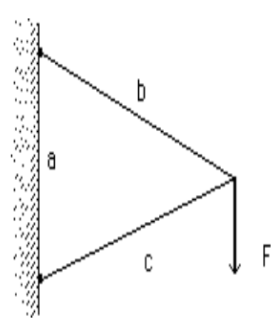


Which variables affect the period of oscillation? Determine a relationship between the period of oscillation and the other variables using dimensional analysis.

2. **Fluid Pressure.** Which quantities determine the overpressure p in a fluid? Derive an expression for p using dimensional analysis.

3. **String Force.** A particle with mass m is attached to a string and moves in a circular path on a frictionless table. The particle's speed is v and the radius of the path is r. The radius is constant. Derive an expression for the force F with which the string affects the particle.

4. **Truss.** A truss is loaded with the force F. The rods with lengths b and c are massless. How large is the tensile force T in the rod with length b?

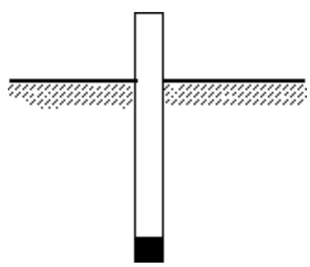


5. How large is the deflection d at the point of application of the force for the truss in problem 4? The rods with lengths b and c have a cross-sectional area A, and the wall is rigid.

6. **Throwing Height.** A stone is thrown straight up with an initial speed v. Determine an expression for the maximum height h the stone reaches. Air resistance is neglected, and the throwing height should depend on gravity. Discuss, based on the result, how high a good high jumper can jump on the moon.

7. **Conical Pendulum.** Which variables determine the period t of a conical pendulum? Derive an expression for t.

8. **Bobbling Buoy.** A cylindrical buoy is equipped with a weight at the lower end so that it floats upright in water. The buoy bobs vertically with a small amplitude. The buoy's diameter is d, its mass is m, and the fluid density is ρ. The buoy is kept in the fluid by gravity. Determine an expression for the bobbing period of the buoy.



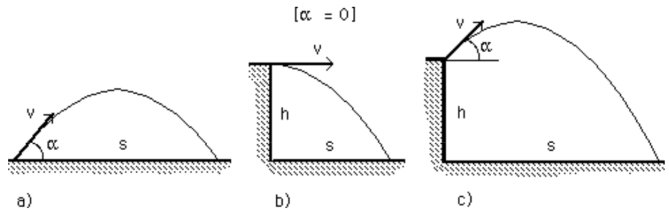
9. **Water Waves.** Determine an expression for how the wave speed v depends on the wavelength λ and other relevant quantities in the following cases. (Since it concerns surface waves, gravity must at least be included.)

   (a) For waves where the water depth d must be considered.

   (b) For waves in deep water, i.e., when $\lambda \ll d$.

10. **Sound Speed.** The speed of sound c in a gas depends on the gas's pressure and density and possibly on their viscosity. Determine c.

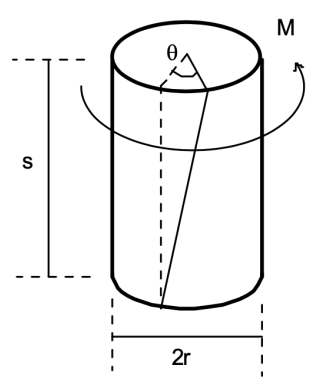
11. **Projectile Range.** A projectile with mass m is fired with an initial speed v and an elevation angle α. Determine, as far as possible, the relationship between the projectile's range s and other quantities in the cases illustrated in the figures below. Air resistance is neglected.



12. **Electrical Problem.** An electric fan with a diameter d requires a certain torque M to rotate at n revolutions per minute. The properties of the air and the design of the fan are significant for M. Determine a dimensionless expression for M.

13. **Torsional Vibration.** A uniform elastic rod with a circular cross-section twists around its axis when it vibrates with the lowest order torsional frequency f. Determine f. The rod's length is s and its radius is r.

14. **Torsion.** When a homogeneous round rod is subjected to a twisting moment M, it deforms as shown in the figure. (It is assumed that the rod is fixed at the lower end.) A measure of the deformation is the angle θ by which the upper end twists.



(a) Determine a relationship between θ and M and other relevant quantities.

(b) How is the relationship simplified by the following information:

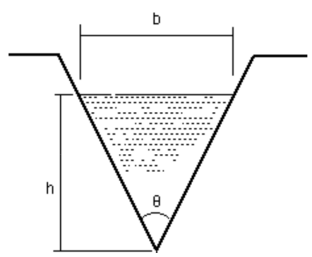
- Since the material in the rod is homogeneous, each unit length must twist equally.
- The angle θ is proportional to the moment M.

15. **A Hydrology Problem.**

    (a) A V-shaped notch is used to measure the flow rate Q (volume per unit time) of a fluid in an open channel. The technique is used, for example, to measure small leakage flows in dams.

    - Q can be expected to depend on:
    - h, fluid level
    - b, free fluid surface width
    - g, gravitational acceleration
    - ρ, fluid density
    - μ, fluid dynamic viscosity

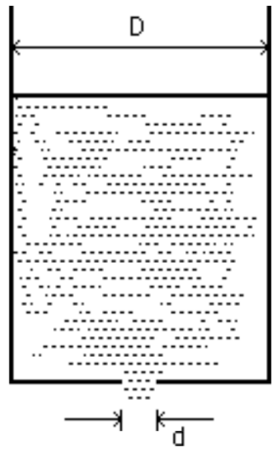
    Determine Q. Three π-groups describe this problem.



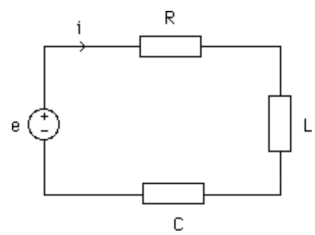
    (b) Calibration of a notch with an opening angle $\theta = 60°$ shows that the flow rate for water is 20 liters/second if $h = 25$ cm. The same notch will be used to measure an oil flow for an oil with a kinematic viscosity ($\nu = \mu/\rho$) 10 times that of water. For which level h is the calibration applicable for the oil? What oil flow rate is obtained for this h value?

16. **Emptying.** A cylindrical vessel is filled with a fluid that can be drained through a hole at the bottom. Investigate how different variables affect the time it takes to empty the container.

17. **Circuit Problem.** An electrical circuit contains a resistance R, an inductance L, a capacitance C, and an electromotive force (emf) e.



The emf is described by $e = at^2$, where a is a constant. What is the current at time t?

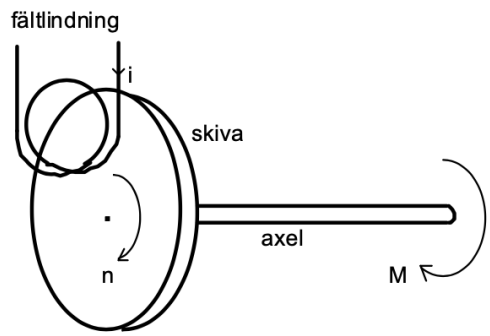


18. **Eddy Current Brake.** The current in the field winding generates a magnetic field, which in turn generates eddy currents in the rotating conductive disk. The interaction between the field current i and the eddy currents produces a braking torque M. The disk's permeability is μ, its resistivity is ρ, and its diameter is d. The rotational speed is n revolutions per minute. Determine M.

Answers to Exercises

In the answers to the exercises in Chapter 5, k is a dimensionless constant. Note that alternative answers exist for the problems marked with an asterisk. This depends on which variables were chosen for elimination (or as base variables).

1 $t = k\sqrt{\frac{s}{g}}$



2 $p = k\rho g h$

3 $F = km\frac{v^2}{r}$

4* $\dfrac{F}{T} = \phi(\frac{b}{a}, \frac{c}{a})$

5* $\dfrac{d}{a} = \phi\left(\frac{a^2}{A}, \frac{b}{a}, \frac{c}{a}, \dfrac{F}{EA}\right)$; a better analysis shows that ϕ is independent of $\frac{a^2}{A}$

6 $h = k\frac{v^2}{g}$

7 $t = \sqrt{\frac{\sigma}{g}}\phi(\alpha)$, $\sigma =$ string length, $\alpha =$ angle between string and vertical line

8* $t = \sqrt{\frac{d}{g}}\phi\left(\dfrac{m}{\rho d^3}\right)$, $d =$ cylinder diameter, $\rho =$ liquids density

9a* $v = \sqrt{\lambda g}\ \phi(\dfrac{d}{\lambda})$

9b $v = k\sqrt{\lambda g}$

10 $v = k\sqrt{\frac{P}{\rho}}$

11a $s = \frac{v^2}{g}\Phi(\alpha)$

11b* $s = h\Phi\left(\frac{v}{\sqrt{gh}}\right)$

11c $s = h\Phi\left(\frac{v}{\sqrt{gh}}, \alpha\right)$ or $s = \frac{v^2}{g}\Phi\left(\frac{gh}{v^2}, \alpha\right)$

12 $M = \rho d^5 n^2 \Phi(R)$, $R =$ ratio of lengths

13* $f^2 s^2 \dfrac{\rho}{G} = \Phi\left(\frac{s}{r}\right)$, it can be seen that Φ is constant

14a* $\theta = \Phi\left(\dfrac{M}{Gr^3}, \dfrac{s}{r}\right)$, $G =$ shear modulus of the rod

14b* $\theta = \frac{s}{r}\Phi\left(\frac{M}{Gr^3}\right)$

14c $\theta = k\frac{Ms}{Gr^4}$

15a e.g., $Q = \sqrt{h^5 g}\ \Phi\left(\dfrac{b}{h}, \dfrac{\rho}{\mu}\sqrt{3gh}\right)$

15b $h = 116\,\text{cm}$, $Q = 928\,\text{liters/s}$

16 Out to the lab and work!

17* $\Phi\left(\dfrac{t}{\sqrt{LC}}, \dfrac{R^2 C}{L}, \dfrac{iR}{a}(R/L)^2\right) = 0$

18* $\Phi\left(\dfrac{n\mu d^2}{\rho}, \dfrac{M}{i^2 \mu d}, R\right) = 0$, $R =$ ratio of lengths

Chapter 6

Measurement Errors

When measuring anything, whether it is time, length, temperature, etc., there is always some degree of error. Some errors are due to the measurement method, others to the instrument, and some to the person measuring.

6.1 Different Types of Errors

Gross Errors

These are simply mistakes in measurement, recording errors, reading errors, forgetting to divide by two, or holding a hand on the scale. The remedies for these errors are orderliness, neatness in the lab, and a splash of coffee for the sleepy researcher.

Systematic Errors

Systematic errors are errors that are repeated in the same way in each measurement. The error can be a ruler with an incorrect scale, an uncalibrated instrument, skewed reading of the scale, etc. These errors are usually constant and, if known, can be subtracted from the measurements. If the errors are not constant, such as a steadily increasing temperature during the experiment, the error becomes harder to detect and also harder to compensate for.

Random Errors

These occur when repeated measurements give different results. The errors are often small and are as often below the mean value as above it. Sources of random errors can be a shaky hand, loose equipment, etc. Random errors are usually nothing to worry about. The remedy is to make many measurements, which will reduce the random error.

It is rare that only one type of error occurs in a measurement series. Usually, the total measurement error is a combination of the three types of errors.

6.2 Many Measurements

Making many measurements can reduce random error. If the measurement error is random, none of the individual measurements can be said to be more "correct" than any other. The best estimate we can make of the true (exact) value is the mean value,

$$\langle x \rangle = \frac{1}{n} \sum_{i=0}^{n} x_i$$

where $\langle x \rangle$ is the mean value of n measured values. The more measurements we make, the closer the mean value will be to the exact value. To describe the measurement series, we need, in addition to the mean value, a measure of how much the mean value deviates from the exact value.

Generally, the measurement result of a measured quantity x is written as

$$x = \langle x \rangle \pm \delta x$$

where

- $\langle x \rangle$ is the approximate value (often the mean value) of x,
- δx is the error margin (often a measure of spread and systematic errors),
- x is the exact value.

The error margin gives an upper limit on how far the mean value is from the exact value. Generally, the error margin often includes a measure of spread, if there are many measurements, together with estimated systematic errors. The measure of spread indicates how much uncertainty the mean

value has due to purely random errors. The systematic error indicates how much uncertainty the mean value has due to, for example, the measurement instrument's error (the instrument's inaccuracy). The measure of spread is added to the systematic error to form the error margin.

In repeated measurements, there is a measure of how much we can expect an individual measurement to deviate from the mean value. This measure is called the standard deviation, s_n,

$$s_n = \left(\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \langle x \rangle)^2 \right)^{1/2}$$

where n is the number of measurements and x_i is the i-th measurement. One might think that by making more measurements, the standard deviation can be reduced, but this is not quite the case. What happens with an increasing number of measurements is that s_n will approach a fixed, specific value that can be said to be the precision of the measurement method. s_n can be large or small and depends entirely on the measurement method used. To reduce the standard deviation, the measurement method should be changed to one with higher precision.

To get a relevant measure of spread for the mean value $\langle x \rangle$, the standard deviation is not enough; we need to go one step further. We assume for simplicity that the errors in the measurements are completely random. If we then make enough measurements, the mean value should become the exact value, and the error in the measurements is reduced to zero. A good estimate of how much the mean value deviates from the exact value is the standard error, S_n,

$$S_n = \frac{s_n}{\sqrt{n}}$$

where n is the number of measurements. We see that regardless of how poor the precision of our measurement method is, the random measurement error can always be reduced to nearly zero if enough measurements are made.

Note that s_n and S_n are two quantities that express different things. s_n is a measure of how much the measurements on average deviate from the mean value, and S_n is a measure of how much the mean value deviates from the exact value.

As a measure of spread in a measurement situation with many measurements, the standard error S_n is used because it gives a good estimate of how far the mean value of the measurements is from the (unknown) exact value.

6.3 Few Measurements

If only one or a few measurements are made, it is not worth calculating the standard deviation and standard error. Instead, one can calculate the maximum deviation from the mean value and use it as a kind of measure of spread. Let's take an example.

6.3.1 Example 1

Imagine measuring the ceiling height in a room (if one wanted to wallpaper with an incredibly expensive wallpaper or something like that). One has a measuring tape available and makes four measurements.

Table 6.1: Results of ceiling height measurement.

Ceiling Height	Measurement (m)
h_1	2.153
h_2	2.154
h_3	2.152
h_4	2.150

The mean value is $\langle h \rangle = 2.15225\,\mathrm{m}$.
And the maximum deviations from the mean value are

$$2.154 - 2.15225 = 0.00175\,\mathrm{m} \quad \text{deviation upwards,}$$

$$2.150 - 2.15225 = -0.00225\,\mathrm{m} \quad \text{deviation downwards.}$$

The maximum deviation (in absolute value) constitutes our error margin. We get the ceiling height

$$h = 2.15225 \pm 0.00225\,\mathrm{m}.$$

To this calculated measure of spread, we add the instrument error. With a measuring tape, we can assume that the inaccuracy is about 1 mm (0.001 m). This instrument error is added to the measure of spread. We get

$$h = 2.15225 \pm 0.00325\,\mathrm{m},$$

which is rounded to

$$h = 2.152 \pm 0.004\,\mathrm{m}.$$

Note how the rounding was done: first, the mean value was rounded. What was "rounded off" is added to the error (i.e., 0.00025 is added to the error, which becomes 0.00350, which is rounded up to 0.004). Errors are always rounded up.

If we had only made one measurement, we would only have one error to consider, namely the instrument error.

6.4 Rounding

When rounding, one usually follows certain simple rules:

- 1, 2, 3, 4 are rounded down,
- 6, 7, 8, 9 are rounded up,
- 5 is rounded as desired (up or down).

Again, remember that *errors are always rounded up.*

6.5 Indirect Measurements

Indirect measurements mean measuring a number of quantities to calculate another via some formula. The formula thus gives an indirectly measured value. Errors in the measured quantities affect the indirectly measured value. Let's take an example:

Example 2

Suppose we want to find the volume V of a sphere by measuring its diameter, D. We make a lot of measurements of D so that we can calculate both the mean value and the error margin. How volume and diameter are related is governed by

$$V = \frac{4}{3}\pi\left(\frac{D}{2}\right)^3 = \frac{\pi}{6}D^3.$$

To calculate the volume V, the mean value of D is used; i.e., $\langle D\rangle$. What we want is

$$V = \langle V\rangle \pm \delta V.$$

What remains to be determined is δV. We use the derivative of the function at the point $D = \langle D \rangle$ and draw the tangent to the curve, see Figure 6.1 below.

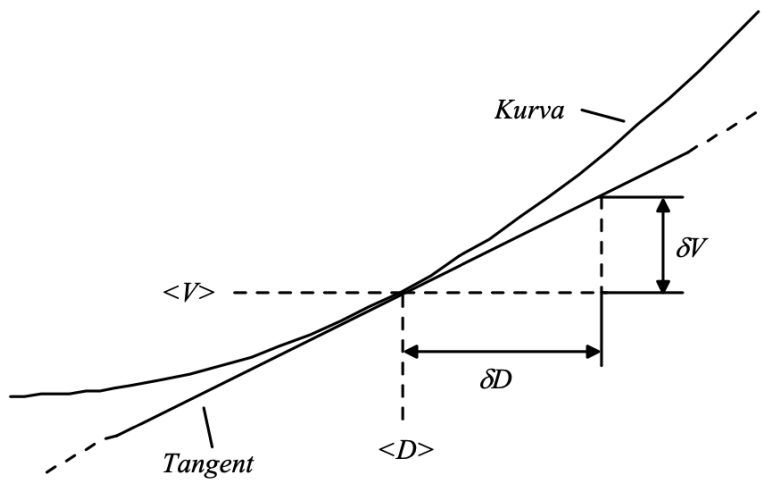


Figure 6.1: The figure shows how the tangent to the curve is used to approximate the error in volume if the error in diameter is known.

A right-angled triangle is formed. One side is the error in D and the other side is the error in V.

By using the fact that the slope of the tangent is the derivative of the function at the point, we can do the following:

$$\frac{\delta V}{\delta D} = \frac{dV}{dD}.$$

The right-hand side is just the derivative of V, so we differentiate V with respect to D:

$$\frac{dV}{dD} = \frac{\pi}{2} D^2.$$

Combining, we get:

$$\delta V = \frac{\pi}{2} D^2 \delta D.$$

But we are only interested in the magnitude of the error (errors can only be added), so we get:

$$\delta V = |\frac{\pi}{2} D^2| \delta D.$$

Note that $|\frac{\pi}{2}D^2|$ is calculated for $D = \langle D \rangle$.

Now we are ready to calculate the error in V, δV, via the error in D, δD.

If there happen to be more variables than one (as in the example above), it is just a matter of adding the error in each variable. If the relationship is

$$G = f(a, b, c, d, \ldots),$$

then the error in G, formally, is

$$\delta G = \left|\frac{\partial f}{\partial a}\right| \delta a + \left|\frac{\partial f}{\partial b}\right| \delta b + \left|\frac{\partial f}{\partial c}\right| \delta c + \ldots.$$

The formula above is called the error propagation formula and it describes how an error in measured quantities propagates through a relationship to the indirectly measured quantity. The "curly" derivatives (∂) function as regular derivatives. By writing, for example, $\frac{\partial f}{\partial a}$, it means that the other variables b, c, etc., are held constant while differentiating f with respect to a.

The error propagation formula works well in most cases, but there are cases where it is unsuitable to use. If the function f has a very steep derivative in some variable, while the error in that variable is large, the linear approximation will create a large gap between the tangent and the curve (see Figure 6.1). This results in the error being simply wrong (!) and thus unusable.

6.6 Presentation of Measurement Data in Diagram Form

Measurement data can be presented both in the form of tables and diagrams. When drawing a graph, one should consider:

1. Choosing the right scales on the axes. The graph should "fill out" the diagram and not just be a small, small line in some corner. Choosing logarithmic scales can be good if the measurement data spans many orders of magnitude. By choosing odd scales, one can deceive a lot, so avoid that.

2. Preferably not cutting the axes and omitting the origin. If possible, choose another scale instead.

3. Measurement values should always be marked with a clear symbol.
4. The axes should always be provided with both symbol and unit.
5. If the measurement values are associated with significant errors, these are drawn as bars around the measured value, see Figure 6.2 below. The total height of the bar corresponds to $2\delta y$, as the error is equally large both above and below the mean value ($\pm\delta y$).

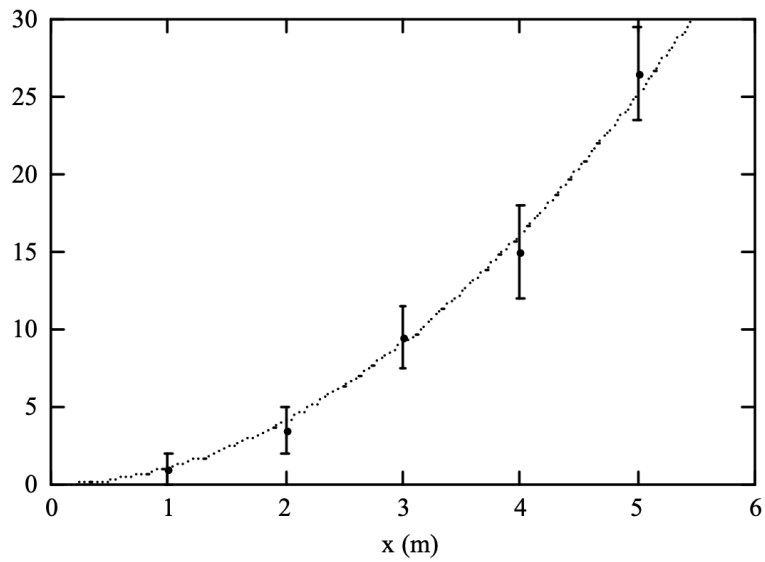


Figure 6.2: The diagram shows an example of how to use error bars. The total height of the bar corresponds to two times the error in v, i.e., $2\delta v$.

Assume that the points in the diagram correspond to measured values. Each value is associated with a certain measurement error. The measurement error for each value is indicated as error bars in the diagram. Using the method in Chapter 2 and the measurement points, we find that the relationship between x and v is

$$v(t) = k \cdot t^2$$

where $[k] = LT^{-3}$. In our case, we find that $k = 1\,\mathrm{m/s^3}$.

The relationship is drawn as the dashed curve in the diagram. It is seen that the curve does not completely fit the measurement values but is clearly within the error limits for each measurement value.

Since the curve lies within the error limits, one can be reasonably sure that the correct relationship has been found. If increased accuracy is desired, the measurement errors must of course be reduced. It is obvious that one strives for as small measurement errors as possible in all experimental activities.

Appendix

A1 Examples of Dimensionless Standard Groups

- Mach number, M:

$$M = \frac{v}{c} \quad \text{(speed/speed of sound)}$$

- Grashof number, Gr:

$$Gr = \frac{s^3 \rho g \gamma \Delta T}{\mu^2} \quad \text{(Inertia force/viscous force in free convection)}$$

- Reynolds number, Re:

$$Re = \frac{va\rho}{\mu} \quad \text{(Inertia force/viscous force in the flow of gases and liquids)}$$

- Nusselt number, Nu:

$$Nu = \frac{hs}{k} \quad \text{(Convective heat transfer/conductive heat transfer)}$$

A2 Definition of Symbols (Mechanics, Heat)

Table 2: Definition of symbols (mechanics, heat)

Physical Quantity	Symbol	Dimension	SI Unit
Density	ρ	$[\rho] = ML^{-3}$	$\{\rho\} = \text{kg m}^{-3}$
Dynamic viscosity	μ	$[\mu] = ML^{-1}T^{-1}$	$\{\mu\} = \text{N s m}^{-2}$
Kinematic viscosity	ν	$[\nu] = L^2T^{-1}$	$\nu = \frac{\mu}{\rho}, \{\nu\} = \text{m}^2\text{s}^{-1}$
Elastic modulus	E	$[E] = ML^{-1}T^{-2}$	$\{E\} = \text{Pa}$
Flow velocity	v	$[v] = LT^{-1}$	$\{v\} = \text{m s}^{-1}$
Speed of sound	c	$[c] = LT^{-1}$	$\{c\} = \text{m s}^{-1}$
Length	a	$[a] = L$	$\{a\} = \text{m}$
Shear modulus	G	$[G] = ML^{-1}T^{-2}$	$\{G\} = \text{Pa}$
Temperature difference	ΔT	$[\Delta T] = \Theta$	$\{\Delta T\} = \text{K}$
Thermal conductivity	k	$[k] = MLT^{-3}\Theta^{-1}$	$\{k\} = \text{W m}^{-1}\text{K}^{-1}$
Gravitational acceleration	g	$[g] = LT^{-2}$	$\{g\} = \text{m s}^{-2}$
Volumetric expansion coefficient	γ	$[\gamma] = 1$	$\{\gamma\} = 1$
Heat transfer coefficient	h	$[h] = MT^{-3}\Theta^{-1}$	$\{h\} = \text{W m}^{-2}\text{K}^{-1}$

A3 Electrical Symbols

Table 3: Electrical symbols

Physical Quantity	Symbol	Dimension	SI Unit
Speed of light in vacuum	c	$[c] = LT^{-1}$	$\{c\} = \mathrm{m/s}$
Permittivity of vacuum	ϵ_0	$[\epsilon_0] = M^{-1}L^{-3}T^4$	$\{\epsilon_0\} = \mathrm{As/Vm}$
Permeability of vacuum	μ_0	$[\mu_0] = MLT^{-2}$	$\{\mu_0\} = \mathrm{Vs/Am}$
Current	i	$[i] = I$	$\{i\} = \mathrm{A}$, Ampere
Charge	q	$[q] = IT$	$\{q\} = \mathrm{As} = \mathrm{C}$ Coulomb
Voltage	u	$[u] = ML^2T^{-3}I$	$\{u\} = \mathrm{V}$ Volt
Resistance	R	$[R] = ML^2T^{-3}$	$\{R\} = \Omega$ Ohm
Inductance	L	$[L] = ML^2T^{-2}$	$\{L\} = \mathrm{Vs/A}$
Capacitance	C	$[C] = M^{-1}L^2T^{-2}$	$\{C\} = \mathrm{As/V} = \mathrm{F}$ Farad
Electric field strength	E	$[E] = MLT^{-3}I$	$\{E\} = \mathrm{V/m}$
Magnetic flux density	B	$[B] = MT^{-2}I$	$\{B\} = \mathrm{T} = \mathrm{Vs/m}^2$ Tesla

A4 Pressure and Temperature

Pressure is defined as force per unit area:

$$p = \frac{F}{A}$$

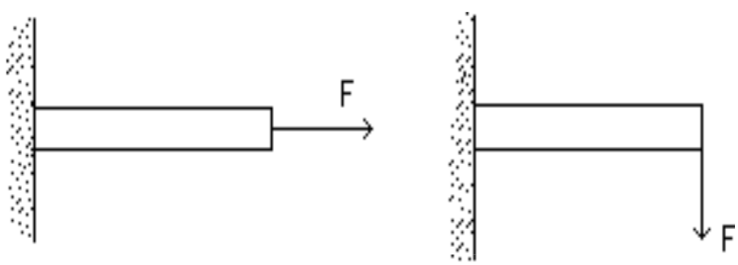
where

$$p = \text{pressure}, \quad F = \text{force}, \quad A = \text{area}.$$

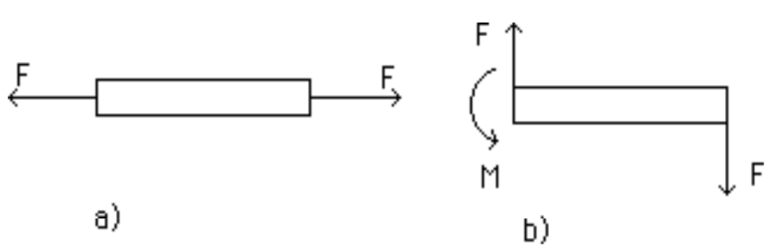
$$[p] = ML^{-1}T^{-2}, \quad [F] = MLT^{-2}, \quad [A] = L^2$$

The force F is considered to be distributed over an area A.

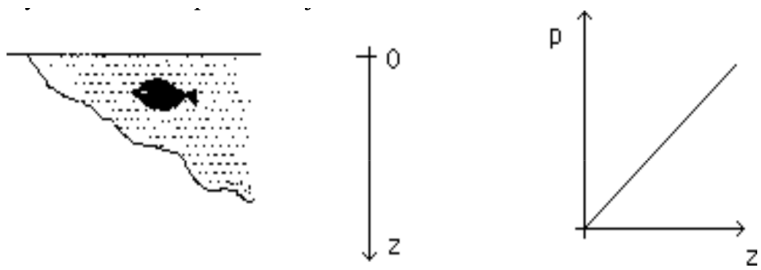
Consider a beam fixed at one end in a wall. The cross-sectional area of the beam is A.



The force F is said to be a normal force in case (a) and a shear force in case (b). Assume the beam does not move. If the weight of the beam mg is much smaller than the force F, a force F from the rest of the beam acts on each part of the beam.



Additionally, in case (b), a moment M acts from the rest of the beam. M corresponds to a normal force that varies over the cross-section with area A. In case (a), the pressure p is called normal stress, and in case (b), it is called



shear stress. Usually, the symbol σ is used for normal stress and τ for shear stress.

In a fluid, the pressure increases with the height of the fluid column.

It holds that:

$$p = \rho g z$$

where
ρ = density of the fluid,
g = gravitational acceleration,
z = coordinate along the vertical line.

The pressure p is set to zero at $z = 0$. Thus, p is the overpressure inside the fluid due to the weight of the fluid. p is equally large in all directions in the fluid. The fluid column must not be too narrow for equation (2) to hold. The width must be large compared to the characteristic length of the water molecules, but do not worry.

Even if you take a very small volume, say... $10^{-9}\,\mathrm{mm}^3$ of a thin gas, such as air at 20°C and normal air pressure, it will contain about 3×10^7 molecules. You realize that equation (2) is fulfilled in almost all cases. Equation (2) can be applied to both liquids and gases. However, the density of a gas is generally about a thousand times less than the density of a liquid. In the general case, the density of the liquid or gas also depends on the pressure. Gases have a significantly higher compressibility than liquids, which are almost incompressible. A good approximation is to assume that the density of a liquid is constant and independent of the pressure. The density of gases varies strongly with pressure (high compressibility). In atmospheric contexts, it can be assumed with good approximation that the density of air is constant and independent of height – if the height differences are less than about 1000 m.

For a gas, the general gas law holds with good approximation:

$$pV = NkT$$

where
V = volume of the gas, N = number of gas molecules in V,
k = Boltzmann's constant, T = absolute temperature.

These quantities have the dimensions:

$$[V] = L^3, \quad [N] = 1, \quad [k] = ML^2T^{-2}\Theta^{-1}, \quad [T] = \Theta.$$

Pressure and temperature are thus proportional quantities in a gas. According to the kinetic theory of gases, both pressure and temperature are functions of the average energy of the gas molecules. Assume now for simplicity that the gas is monoatomic. Then, the following holds (where $\langle ... \rangle$ denotes the average value):

$$p = \frac{2}{3} n \langle \frac{mv^2}{2} \rangle; \quad n = \frac{N}{V}$$

and

$$3kT = \langle \frac{mv^2}{2} \rangle, \quad \langle v^2 \rangle = 3\langle v_x^2 \rangle$$

with

$$v = \text{atomic velocity}, \quad m = \text{atomic mass}, \quad n = \text{number per unit volume}$$

$$[v] = LT^{-1}, \quad [m] = M, \quad [n] = 1.$$

A definition of temperature that also applies to solids and liquids is desirable. For this purpose, it is necessary to use thermodynamic concepts. Consider a reversible process where an ideal heat engine takes heat Q_1 from a reservoir whose absolute temperature is T_1 and releases heat Q_2 to a reservoir whose temperature is T_2. The engine performs the mechanical work

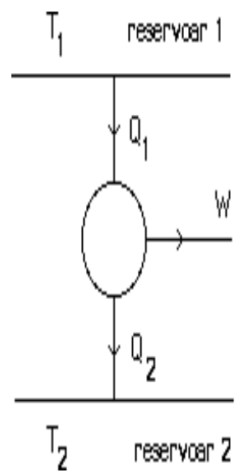
$$W = Q_2 - Q_1.$$

According to the second law of thermodynamics, it follows that

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

Furthermore,

$$[Q_1] = [Q_2] = [W] = [\text{energy}] = ML^2T^{-2}$$



The efficiency of a reversible process is the highest possible and is independent of the heat engine and heat reservoirs. Two ideal heat engines connected in series are equivalent to a single engine. With the help of these facts, it is possible to define a temperature scale, the so-called thermodynamic or absolute temperature scale. This scale is equivalent for a gas to the scale obtained via the kinetic theory of gases.

The dimension length has a base unit, and every length is a multiple of this unit. Length is a linear quantity, and only linear quantities are useful in dimensional analysis. The absolute temperature T with dimension Θ can also be considered a linear quantity in problems involving the conversion of heat to work and vice versa, heat conduction, and thermal expansion of solids. Thus, the dimensions M, L, T, and Θ are used in thermodynamics.

A5 Methods for Planning and Conducting an Experiment

These methods are examples of how to conduct an experiment.

Method 1

1. Identify the variables affecting the experiment:

   - Dependent variable y
   - Independent variables $x, z, h, p, q, \ldots$

2. Guess a relationship between the variables. A good starting point is often to use a factorial power function as an approach:

$$y = C \cdot x^{\alpha} \cdot z^{\beta} \cdot h^{\gamma} \cdot p^{\delta} \cdot q^{\epsilon} \ldots$$

   where C should be dimensionless.

3. Determine how many dimensions are involved using dimensional analysis. (Often 3 dimensions: L, T, M). This means that 3 unknown exponents can be determined using dimensional analysis without conducting any experiments. It is good to determine only 2 exponents using dimensional analysis to leave one relationship for verification.

4. Keep all independent variables constant except one. Write the approach as, for example:

$$y = C_1 \cdot x^{\alpha},$$

   where $C_1 = C \cdot z^{\beta} \cdot h^{\gamma} \cdot p^{\delta} \cdot q^{\epsilon} \ldots$

   Conduct measurements, record in a value table, plot diagrams, linearize, and determine α. Use at least 4 measurement values for each diagram.

   Note: If the measurement points (after linearizing $y = C \cdot x^{\alpha}$) do not lie on a straight line, the assumption $y = C \cdot x^{\alpha}$ is incorrect. Make a new assumption using the various graphs. Perhaps the curve is composed of two terms where one term dominates for small x-values while the other dominates for large x-values; or the curve does not pass through the origin; or it is an exponential function; or ...

5. Continue to experimentally determine the other independent variables by varying one variable at a time in the same way. For example, if you have:

$$y = C \cdot x^{\alpha} \cdot z^{\beta} \cdot h^{\gamma} \cdot p^{\delta} \cdot q^{\epsilon},$$

   you can continue to determine β:

$$y = C_2 \cdot z^{\beta},$$

   where $C_2 = C \cdot x^{\alpha} \cdot h^{\gamma} \cdot p^{\delta} \cdot q^{\epsilon} \ldots$

Note: Here, you can also do as in "A Special Technique" on page 11 of the compendium (especially useful if you cannot get enough measurement values otherwise). Since α is now known, form:

$$\frac{y}{x^\alpha} = C \cdot z^\beta \cdot h^\gamma \cdot p^\delta \cdot q^\epsilon,$$

$$\frac{y}{x^\alpha} = C_3 \cdot z^\beta,$$

where $C_3 = h^\gamma \cdot p^\delta \cdot q^\epsilon \ldots$

When linearizing, plot $\ln\left(\frac{y}{x^\alpha}\right)$ on the vertical axis and $\ln(z)$ on the horizontal axis.

Determine all exponents except 2 (possibly 3). These can be determined using dimensional analysis.

6. Finally, determine the dimensionless constant C. A good way to do this is to plot y as a function of $x^\alpha \cdot z^\beta \cdot h^\gamma \cdot p^\delta \cdot q^\epsilon$. This also provides a good check that it is correct.

Method 2: π-Group Analysis

1. Make a detailed sketch of the experimental setup. Think about how the experiments should be conducted.

2. Consider which quantities can affect the experiment.

3. Can any quantity be neglected? If you suspect this, conduct experiments to see if the suspicion is correct.

4. Determine the dependent variable.

5. Then choose the independent variables. Here, variables that cannot always be varied should also be included, e.g., g, E, G, etc.

   It is usually advantageous to choose the independent variables as simple as possible, e.g., instead of the area, choose the radius or length and width.

   Do not introduce a variable twice, e.g., if you have chosen mass and volume, do not also choose density as it is already included as a variable.

   Create a table:

Quantity	Symbol	Dimension
Dependent variable Independent variables		

6. Determine the number of π-groups. The number is normally (Total number of variables, n) - (Total number of dimensions involved, m), i.e., $n-m$. Under certain circumstances, there may be fewer π-groups, but we will not go into that here.

   Determine what the π-groups will be according to the textbook (primarily the "striking method"). Note: π_1 should contain the dependent variable.

7. Guess a relationship between the π-groups. Normally, it is assumed:

$$\pi_1 = C \cdot \pi_2^{\alpha} \cdot \pi_3^{\beta} \cdot \pi_4^{\gamma} \cdot \pi_5^{\delta} \ldots$$

   Keep, for example, $\pi_3, \pi_4, \pi_5 \ldots$ constant. You then get:

$$\pi_1 = C_1 \cdot \pi_2^{\alpha},$$

   where $C_1 = C \cdot \pi_3^{\beta} \cdot \pi_4^{\gamma} \cdot \pi_5^{\delta} \ldots$

   Conduct experiments. Plot a graph with π_1 on the "y-axis" and π_2 on the "x-axis". Is the assumption $\pi_1 = C_1 \cdot \pi_2^{\alpha}$ reasonable? If so, try to determine α by linearizing.

   Then determine the other exponents by conducting new experiments and then linearizing. (Here, all exponents must be determined experimentally, unlike Method 1).

8. Finally, determine the constant C by linearizing, and summarize the final formula.

A6 Literature List

The list below is intended as an introduction to the literature on the subject. There is a large amount of literature available, and only a few books are listed below. If you want to know more about dimensional analysis for a specific problem or problem area, it may be worthwhile to look at the books that deal specifically with that area. For example, textbooks in fluid mechanics usually contain some material on dimensional analysis for flow problems.

First, two generally held books

- Jönsson, B., Reistad N., *Experimentell fysik*, Second edition, Studentlitteratur, Lund, 1987.
  A book that is easy and fun to read. Besides containing good physics, it also includes some history and curiosities.

- Squires, G.L., *Practical physics*, 3rd edition, Cambridge University Press, Cambridge, 1985.
  A good handbook for physical experiments, interpretation and treatment of measurement data, and report writing.

Then, some books specialized in dimensional analysis

The books cover many aspects of dimensional analysis. They are advanced by our standards but are definitely worth reading if you want to delve deeper into dimensional analysis, π-groups, and geometric and dynamic similarity.

- Sedov, L.I., *Similarity and dimensional methods in mechanics*, 10th edition, CRC Press, 1993.

- Taylor, E.S., *Dimensional analysis for engineers*, Clarendon Press, Oxford, 1974.

- Isaacson, E. de st Q., Isaacson, M. de st Q., *Dimensional methods in engineering and physics*, John Wiley & Sons, New York, 1975.